

Axonal distribution of mitochondria maintains neuronal autophagy during aging via eIF2 β

Reviewed Preprint

v4 • October 31, 2025

Revised by authors

Reviewed Preprint

v3 • September 12, 2025

Reviewed Preprint

v2 • January 23, 2025

Reviewed Preprint

v1 • March 25, 2024

Kanako Shinno, Yuri Miura, Koichi M Iijima, Emiko Suzuki, Kanae Ando 

Department of Biological Sciences, Graduate School of Science, Tokyo Metropolitan University, Tokyo, Japan • Research Team for Mechanism of Aging, Tokyo Metropolitan Institute for Geriatrics and Gerontology, Tokyo, Japan • Department of Alzheimer's Disease Research, National Center for Geriatrics and Gerontology, Ōbu, Japan • Department of Experimental Gerontology, Graduate School of Pharmaceutical Sciences, Nagoya City University, Nagoya, Japan • Gene Network Laboratory, National Institute of Genetics and Department of Genetics, SOKENDAI, Mishima, Japan • Department of Biological Sciences, School of Science, Tokyo Metropolitan University, Tokyo, Japan

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

eLife Assessment

In flies defective for axonal transport of mitochondria, the authors report the upregulation of one subunit, the beta subunit, of the heterotrimeric eIF2 complex via mass spectroscopy proteomics. Neuronal overexpression of eIF2 β phenocopied aspects of neuronal dysfunction observed when axonal transport of mitochondria was compromised. Conversely, lowering eIF2 β expression suppressed aspects of neuronal dysfunction. While these are intriguing and **useful** observations, technical weaknesses limit the interpretation. On balance, the evidence supporting the current claims is suggestive but **incomplete**, especially concerning the characterization of the eIF2 heterotrimer and the data regarding translational regulation.

<https://doi.org/10.7554/eLife.95576.4.sa3>

Abstract

Neuronal aging and neurodegenerative diseases are accompanied by proteostasis collapse, while the cellular factors that trigger it have not been identified. Impaired mitochondrial transport in the axon is another feature of aging and neurodegenerative diseases. Using *Drosophila*, we found that genetic depletion of axonal mitochondria causes dysregulation of protein degradation. Axons with mitochondrial depletion showed abnormal protein accumulation and autophagic defects. Lowering neuronal ATP levels by blocking glycolysis did not reduce autophagy, suggesting that autophagic defects are associated with mitochondrial distribution. We found that eIF2 β was increased by the depletion of axonal mitochondria via proteome analysis. Phosphorylation of eIF2 α , another subunit of eIF2, was lowered, and global translation was suppressed. Neuronal overexpression of eIF2 β phenocopied the autophagic defects and neuronal dysfunctions, and lowering eIF2 β expression rescued those perturbations caused by depletion of axonal mitochondria. These results indicate the mitochondria-eIF2 axis maintains proteostasis in the axon, of which

disruption may underlie the onset and progression of age-related neurodegenerative diseases.

Highlights

- Loss of axonal mitochondria impairs autophagy and accumulates proteins in the axon
- Loss of axonal mitochondria increases eIF2 β and decreases p-eIF2 α
- Neuronal upregulation of eIF2 β induces autophagic defects and locomotor dysfunction
- Lowering eIF2 β rescues autophagic defects caused by loss of axonal mitochondria

Introduction

Neurons have a morphologically complex architecture composed of microcompartments and require tight regulation of the abundance of proteins and organelles spatially and temporally¹. Such control of protein amounts, or proteostasis, is essential for neuronal functions² and is achieved through the orchestration of protein expression, folding, trafficking, and degradation controlled by intrinsic and environmental signals³. Translation is initiated by the eukaryotic initiation factor 2 (eIF2) complex⁴. eIF2, a heterotrimer of α , β , and γ subunits, transports Met-tRNA to the ribosome in a GTP-dependent manner⁵. Under stressed conditions, phosphorylation of eIF2 α attenuates global translation and initiates translation of mRNAs related to the integrated stress response (ISR)⁶. As for protein degradation, autophagy and proteasome are major systems that maintain proteostasis⁷. The proteasome degrades unnecessary proteins, followed by regulated ubiquitination processes⁸, and autophagy removes damaged or harmful components, including large protein aggregates and organelles, through catabolism (selective autophagy)⁹. In addition to autophagy induced by acute stressors, a basal level of selective autophagy mediates the global turnover of damaged proteins¹⁰.

Such constitutive autophagy decreases during aging, which may underlie declines in the structural and functional integrity of neurons¹¹. Decreased protein degradation and accumulation of abnormal proteins also contribute to increased risks of neurodegenerative diseases. Age-related neurodegenerative diseases such as Alzheimer's disease and Parkinson's disease are often associated with the accumulation of misfolded proteins such as amyloid- β , tau, and α -synuclein¹². Enhancement of autophagy mitigates age-related dysfunctions and neurodegeneration caused by proteotoxic stress¹³. However, it is not fully understood how aging disrupts the regulation of this constitutive autophagy.

Neurons are also highly energy-demanding cells. At nerve terminals, action potentials trigger the release of neurotransmitters via exocytosis of synaptic vesicles, which requires a constant supply of ATP and calcium buffering¹⁴. Such neuronal activity relies on mitochondrial functions¹⁵, and mitochondria are actively transported from their major sites of biogenesis in soma to axons¹⁶. However, the axonal transport of mitochondria declines during aging^{17,18,19}. Reduced axonal transport of mitochondria is thought to contribute to age-related declines in neuronal functions^{17,19,20,21}. The number of functional mitochondria in synapses is reduced in the brains of patients suffering from age-related neurodegenerative diseases such as Alzheimer's disease²², and mutations in genes involved in mitochondrial dynamics are linked to neurodegenerative diseases²³. The mislocalization of mitochondria is sufficient to cause age-dependent neurodegeneration in *Drosophila* and mice^{24,25}, indicating that the proper

distribution of mitochondria is essential to maintain neuronal functions. Thus, depletion of functional mitochondria from axons and proteostasis collapse are common features of aging and neurodegenerative diseases.

Mitochondrial transport is regulated by a series of molecular adaptors that mediate the attachment of mitochondria to molecular motors¹⁶. In *Drosophila*, mitochondrial transport is mediated by *milton* and *Miro*, which attaches mitochondria to microtubules via kinesin heavy chain^{26,27}. In the absence of *milton* or *Miro*, synaptic terminals and axons lack mitochondria, although mitochondria are numerous in the neuronal cell body²⁸. We previously reported that RNAi-mediated knockdown of *milton* or *Miro* in neurons causes a reduction in axonal mitochondria, age-dependent locomotor defects²⁹, and age-dependent neurodegeneration in neuropile area starting around 30 days after eclosion (day-old)²⁴, and enhances axon degeneration caused by human tau proteins²⁴, suggesting that these flies can be used as a model to analyze the effect of depletion of axonal mitochondria during aging. In this study, we investigated a causal relationship between mitochondrial distribution and neuronal proteostasis by using neuronal knockdown of *milton*. We found that depletion of axonal mitochondria reduced autophagy and increased the accumulation of aggregated proteins in the axon prior to gross neurodegeneration. Proteome analysis and follow-up biochemical analyses revealed that neuronal knockdown of *milton* increased eIF2 β levels and lowered phosphorylation of eIF2 α in the axon. In addition, *milton* knockdown suppressed global translation. Overexpression of eIF2 β was sufficient to decrease autophagy and induce neuronal dysfunction, and genetic suppression of eIF2 β restored autophagy and improved neuronal function in the *milton* knockdown background. These findings suggest that loss of axonal mitochondria and elevated levels of eIF2 β mediate proteostasis collapse and neuronal dysfunction during aging.

Results

Depletion of axonal mitochondria by knockdown of *milton* or *Miro* causes protein accumulation in the axon

In *Drosophila*, mitochondrial transport is mediated by *milton* and *Miro*, which attach mitochondria to microtubules via kinesin heavy chain^{26,27} (Figure 1A). It has been reported that expression of *milton* RNAi in neurons via pan-neuronal elav-GAL4 driver reduced *milton* protein levels in *Drosophila* head lysate to 40% and mito-GFP signals in axons to 50%^{24,29}.

To test how loss of axonal mitochondria affects proteostasis in neurons, we first examined the accumulation of ubiquitinated proteins. At 14-day-old, more ubiquitinated proteins were deposited in the brains of *milton* knockdown flies than in those of age-matched control flies (Figure 1B, $p < 0.005$ between control RNAi and *milton* RNAi). There was no significant increase in ubiquitinated proteins in *milton* knockdown flies at 1-day old, suggesting that the accumulation of ubiquitinated proteins caused by *milton* knockdown is age-dependent (Figure S1). We also analyzed the effect of the neuronal knockdown of *Miro*, a partner of *milton*, on the accumulation of ubiquitin-positive proteins. Since severe knockdown of *Miro* in neurons causes lethality, we used UAS-*Miro* RNAi strain with low knockdown efficiency, whose expression driven by elav-GAL4 caused 30% reduction of *Miro* mRNA in head extract²⁴. Although there was a tendency for increased ubiquitin-positive puncta in *Miro* knockdown brains, the difference was not significant (Figure 1B, $p > 0.05$ between control RNAi and *Miro* RNAi). These data suggest that the depletion of axonal mitochondria induced by *milton* knockdown leads to the accumulation of ubiquitinated proteins before neurodegeneration occurs.

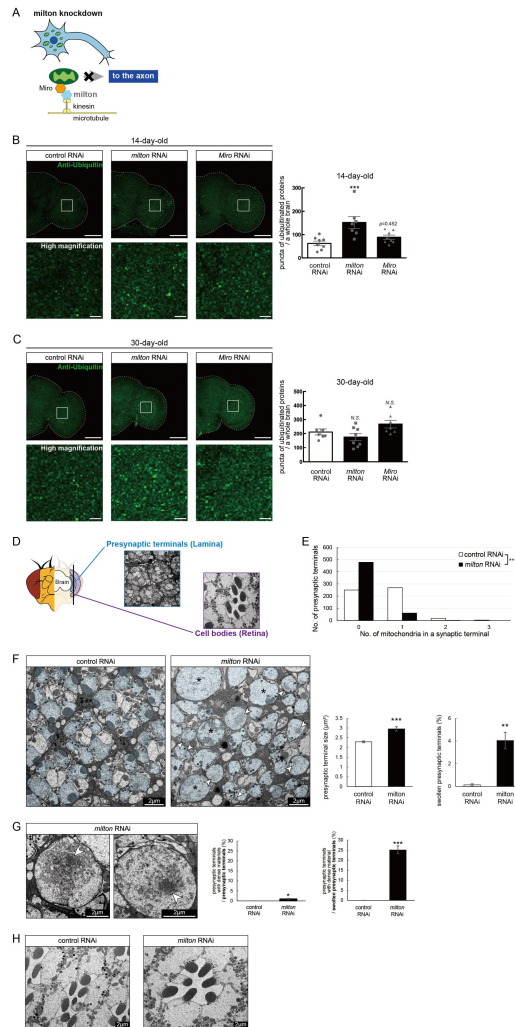


Figure 1.

Knockdown of *milton* or *Miro* causes protein accumulation in the axon

(A) Schematic representation of the mitochondrial transport machinery. Knockdown of *milton*, an adapter protein for mitochondrial transport, depletes mitochondria in the axon. (B, C) Ubiquitinated proteins in brains with neuronal knockdown of *milton* or *Miro*. Brains dissected at 14-day-old (B) or 30-day-old (C) were immunostained with an antibody against ubiquitin. Firefly *luciferase* RNAi was used as a control. Representative images (left) and quantitation of the number of ubiquitin-positive puncta (right) are shown. Scale bars of hemibrains, 100 μ m, Scale bars of high magnifications, 10 μ m. Means $\pm$ SE, $n = 8$. *N.S.*, $p > 0.05$; $***p < 0.005$ (one-way analysis of variance (ANOVA) followed by Tukey's honestly significant difference (HSD) *post hoc* test). (D) Cross sections in the lamina and in the retina were used to analyze the ultrastructure of synapses and cell bodies, respectively. *milton* RNAi was expressed in the retina and neurons via a combination of GAL4 drivers, a pan-retinal *gmr*-GAL4 and pan-neuronal *elav*-GAL4. (E) Quantitation of the number of mitochondria in a presynaptic terminal from transmission electron micrographs. 180 presynaptic terminals from cross-sections of the lamina from three brains were analyzed. $***p < 0.005$ (Chi-square test). (F, G) Presynaptic terminals of photoreceptor neurons of control and *milton* knockdown flies. Photoreceptor neurons are highlighted in blue. Swollen presynaptic terminals (asterisks in (F)), characterized by the enlargement and higher circularity, were found more frequently in *milton* knockdown neurons. Arrowheads indicate presynaptic terminals with dense materials. Scale bars, 2 μ m. Representative images (Left) and quantitation (Right) are shown. 918–1118 from three heads were quantified for the percentage of swollen presynaptic terminals, and 180 presynaptic terminals from three heads were quantified for the size of presynaptic terminals. Mean $\pm$ SE, $**p < 0.01$, $***p < 0.005$ (Student's *t*-test). (G) Dense materials (arrowheads in (G)) in the presynaptic terminals of *milton* knockdown neurons. Scale bars, 2 μ m. The ratio of presynaptic terminals containing dense materials were quantified from 918–1118 presynaptic terminals from three heads. Mean $\pm$ SE, $***p < 0.005$ (Student's *t*-test). (H) Cell bodies of photoreceptor neurons of control and *milton* knockdown flies. Scale bars, 2 μ m. Flies were 27-day-old.

It has been reported that ubiquitinated proteins accumulate with aging³⁰; thus, we analyzed the accumulation of ubiquitinated proteins in aged brains (30-day-old) with *milton* knockdown. The number of puncta of ubiquitinated proteins did not significantly differ between control and *milton* knockdown flies or between control and *Miro* knockdown flies (**Figure 1C**, $p > 0.05$). These results suggest that depletion of axonal mitochondria may have more impact on proteostasis in young neurons than in old neurons.

We examined the ultrastructure of presynaptic terminals and cell bodies in photoreceptor neurons with *milton* knockdown by transmission electron microscopy in 27-day-old flies (**Figure 1D**). As previously reported²⁴, the number of mitochondria in presynaptic terminals decreased in *milton* knockdown (**Figure 1E**). The swelling of presynaptic terminals, characterized by the enlargement and roundness, was not reported at 3-day-old²⁴ but observed at this age with about 4% of total presynaptic terminals (**Figure 1F**, asterisks).

Some presynaptic terminals of *milton* knockdown neurons contained dense materials (**Figure 1F and G**, arrowheads). Dense materials are rarely found in age-matched control neurons, indicating that *milton* knockdown induces abnormal protein accumulation in the presynaptic terminals (**Figure 1G and H**). In *milton* knockdown neurons, dense materials are found in swollen presynaptic terminals more often than in presynaptic terminals without swelling, suggesting a positive correlation between the disruption of proteostasis and axonal damage (**Figure 1G**). In contrast, dense materials were not observed in cell bodies in the *milton* knockdown retina (**Figure 1H**). These results indicate that the depletion of axonal mitochondria induces protein accumulation in the axon.

Depletion of axonal mitochondria impairs protein degradation pathways

Since abnormal proteins were accumulated in *milton* knockdown brains, we next examined if protein degradation pathways were suppressed. We analyzed autophagy via western blotting of the autophagy markers LC3 and p62³¹. During autophagy progression, LC3 is conjugated with phosphatidylethanolamine to form LC3-II, which localizes to isolation membranes and autophagosomes. LC3-I accumulation occurs when autophagosome formation is impaired, and LC3-II accumulation is associated with lysosomal defects^{31,32}. p62 is an autophagy substrate, and its accumulation suggests autophagic defects^{31,32}. We found that *milton* knockdown increased LC3-I, and the LC3-II/LC3-I ratio was lower in *milton* knockdown flies than in control flies at 14-day-old (**Figure 2A**). We also analyzed p62 levels in head lysates sequentially extracted using detergents with different stringencies (1% Triton X-100 and 2% SDS). Western blotting revealed that p62 levels were increased in the brains of 14-day-old of *milton* knockdown flies (**Figure 2B**). The increase in the p62 level was significant in the Triton X-100-soluble fraction but not in the SDS-soluble fraction (**Figure 2B**), suggesting that depletion of axonal mitochondria impairs the degradation of less-aggregated proteins. Proteasome activity was also significantly decreased in brains with neuronal knockdown of *milton* (**Figure 2C**, $p < 0.005$).

At 30-day-old, LC3-I was still higher, and the LC3-II/LC3-I ratio was lower, in *milton* knockdown compared to the control (**Figure 2D**). At this age, *milton* knockdown increased p62 significantly in 1% Triton X-100 fraction and 2% SDS fraction (**Figure 2E**). Proteasome activities were also decreased in *milton* knockdown flies at 30-day-old (**Figure 2F**). These results indicate that depletion of axonal mitochondria impairs protein degradation pathways.

ATP deprivation does not impair autophagy

milton knockdown downregulates ATP in the axon³³. To examine whether the disruption of protein degradation pathways by *milton* knockdown is due to ATP deprivation, we investigated the effects of knocking down phosphofructokinase (*Pfk*), a rate-limiting enzyme in glycolysis, on protein degradation pathways. Neuronal knockdown of *Pfk* was reported to lower ATP levels in

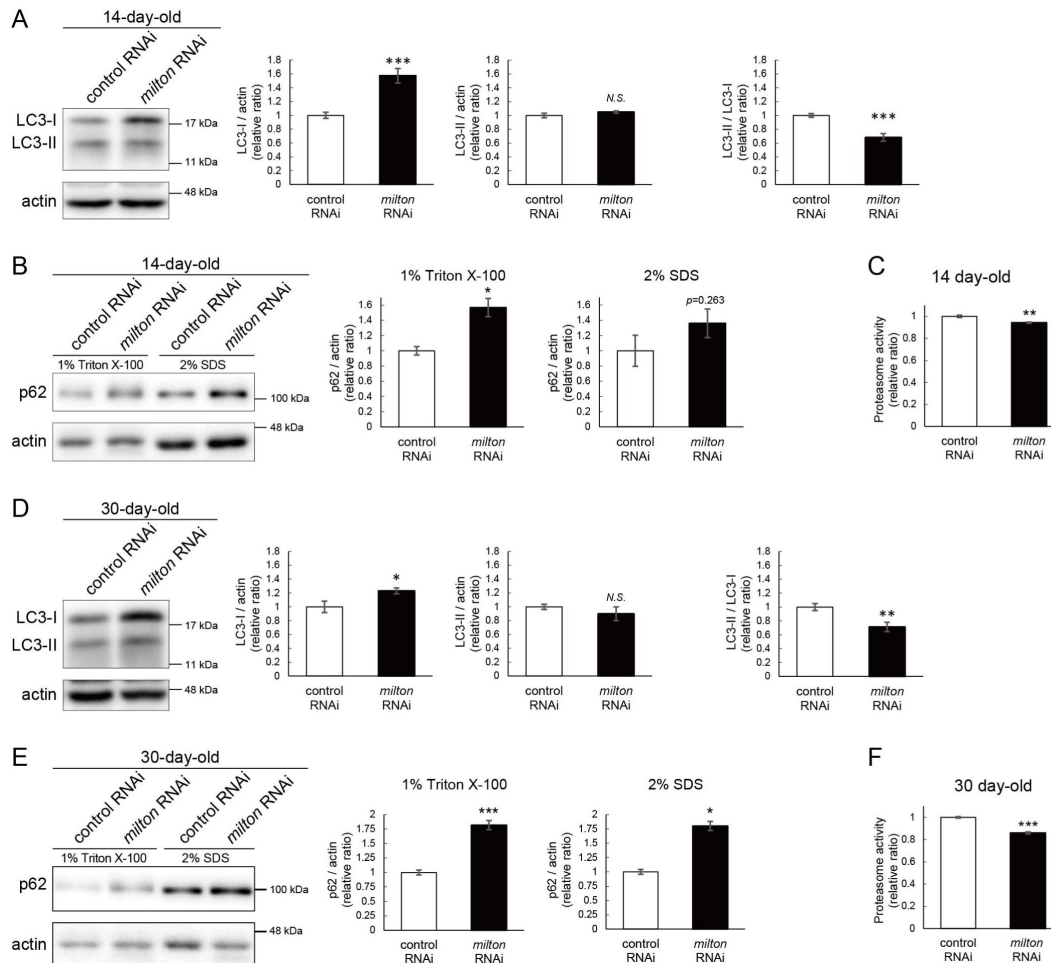


Figure 2.

***milton* knockdown impairs protein degradation pathways**

(A, B) Western blotting of head extracts of control and *milton* knockdown flies with antibodies against LC3 (A) and Ref2P, the fly homolog of mammalian p62 (B). For the analyses of p62 levels, heads were extracted with 1% Triton X-100 or 2% SDS (B). Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Actin was used as a loading control. Means $\pm$ SE, $n = 6$ (LC3), $n = 3$ (p62). (C) Proteasome activity in head extracts of control and *milton* knockdown flies was measured by hydrolysis of Suc-LLVY-AMC at 14-day-old. Means $\pm$ SE, $n = 3$. (D, E) Western blotting of head extracts of 30-day-old control and *milton* knockdown flies. Blotting was performed with anti-LC3 (D) and anti-p62 (E) antibodies. Representative blots (left) and quantitation (right) are shown. Actin was used as a loading control. Means $\pm$ SE, $n = 6$ (LC3), $n = 3$ (p62). (F) Proteasome activity in head extracts of 30-day-old control and *milton* knockdown flies. Means $\pm$ SE, $n = 3$. N.S., $p > 0.05$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.005$ (Student's *t*-test).

brain neurons³³, *Pfk* knockdown and *milton* knockdown decreased ATP to similar levels (**Figure 3A-C**). However, in contrast with *milton* knockdown, *Pfk* knockdown did not affect the levels of LC3-I, LC3-II or the LC3-II/LC3-I ratio (**Figure 3D**). *Pfk* knockdown decreased p62 level (**Figure 3E**), suggesting that autophagy is promoted. On the other hand, proteasome activity was decreased by *Pfk* knockdown (**Figure 3F**). These results suggest that the downregulation of axonal ATP upon depletion of axonal mitochondria decreases proteasome activity, but not autophagy.

Proteome analysis suggests that depletion of axonal mitochondria causes disruption of autophagy and premature aging

To identify the pathways that mediate the decrease in autophagy in *milton* knockdown brains, we performed proteome analysis to systematically detect differentially expressed proteins upon neuronal knockdown of *milton*. We analyzed flies at 7- and 21-day-old, the age before autophagic defects are detected and the age just before the onset of neurodegeneration, respectively (**Figure 4A**). 1039 proteins were detected by liquid chromatography-tandem mass spectrometry (LC-MS/MS). Expression of 36 proteins was significantly increased (22 proteins) or decreased (14 proteins) by *milton* knockdown at 7-day-old (**Figure 4B**, Tables 1 and S1). At 21-day-old, the expression of 41 proteins (31 upregulated and ten downregulated proteins) was significantly altered in *milton* knockdown flies compared with control flies (**Figure 4C**, Tables 1 and S1). The “Interaction search” algorithm using KeyMolnet showed that proteins whose expression was significantly altered in the brains of *milton* knockdown flies at both 7- and 21-day-old were closely associated with the autophagic pathway (Table 2). Proteins involved in pathways characteristics of aging, such as the immune response (Transcriptional regulation by STAT), cancer (Transcriptional regulation by SMAD, Transcriptional regulation by myc), longevity (Transcriptional regulation by FOXO, Sirtuin signaling pathway), and stress responses (HSP90 signaling pathway, MAPK signaling pathway)^{34,35}, were enriched in the proteome profiles of *milton* knockdown flies compared with those of control flies at 7-day-old (Table 2), suggesting that depletion of axonal mitochondria accelerates aging in the brain.

Depletion of axonal mitochondria upregulates eIF2 β and decreases phosphorylation of eIF2 α

Differentially expressed proteins at 7-day-old flies may reflect alterations that are causal for autophagic defects. We noticed that the expression level of eIF2 β was 2.465-fold higher in the brains of *milton* knockdown flies than in those of control flies (**Figure 4B and D**). Upregulation of eIF2 β in the brains of *milton* knockdown flies was confirmed by western blotting. *milton* knockdown increased eIF2 β protein levels more than twice (**Figure 4E**), but did not change the level of *eIF2 β* mRNA (**Figure 4F**).

We also investigated age-dependent changes in eIF2 β by western blotting of control flies at 7-, 21-, 35-, and 49-, and 63-day-old. eIF2 β levels increased during aging until 49-day-old (**Figure 4G**). These results suggest that upregulation of eIF2 β in *milton* knockdown fly brain reflects early an onset of age-dependent increase of eIF2 β levels. eIF2 β is a subunit of the eukaryotic initiation factor 2 (eIF2) complex, which is critical for translation initiation and the integrated stress response (ISR)⁴. eIF2 is a heterotrimer of α , β , and γ subunits, and eIF2 α is phosphorylated during the ISR⁶. As for the other subunits of eIF2 complex, proteome analysis did not detect a significant difference in the protein levels of eIF2 α and eIF2 γ between *milton* knockdown and control flies at 7- and 21-day-old (**Figure 4D**). Western blotting of brain lysates showed that *milton* knockdown reduced eIF2 α levels (**Figure 5A**), while p-eIF2 α levels were not significantly affected (**Figure 5B**).

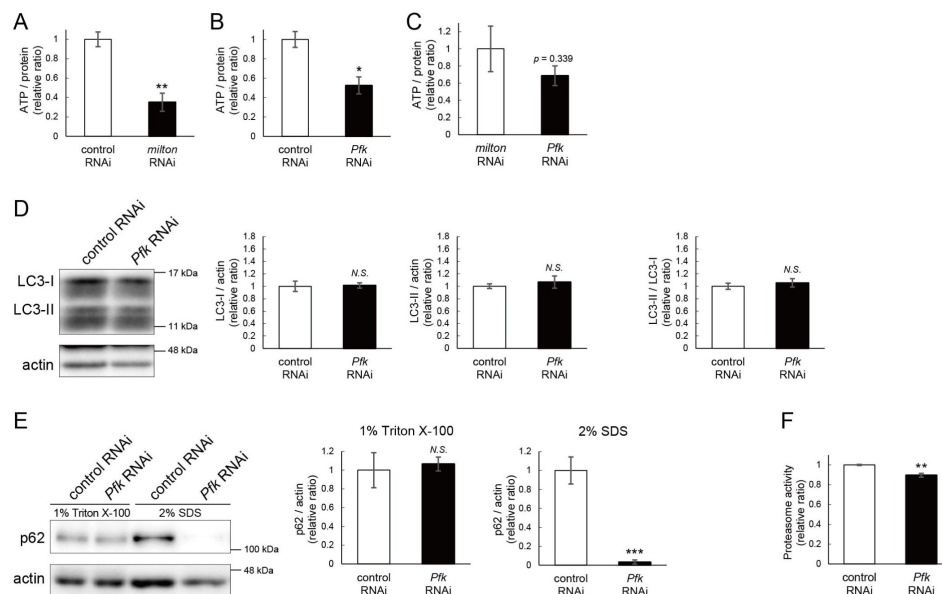


Figure 3.

ATP deprivation does not impair autophagy

(A-C) ATP levels in brain extracts of control and *milton* knockdown flies (A) and control and *Pfk* knockdown flies (B) and comparison of the effects of *milton* knockdown and *Pfk* knockdown on ATP levels (C). Flies were 14-day-old. Means $\pm$ SE, $n = 3$. (D, E) Western blotting of head extracts of flies with neuronal expression of control or *Pfk* RNAi. Blotting was performed with anti-LC3 (D) and anti-p62 (E) antibodies. For analyses of p62 levels, heads were extracted with 1% Triton X-100 or 2% SDS. Representative blots (left) and quantitation (right) are shown. Actin was used as a loading control. Means $\pm$ SE, $n = 6$ (LC3), $n = 3$ (p62). (F) Proteasome activity in head lysates of flies with neuronal expression of control or *Pfk* RNAi was measured by hydrolysis of Suc-LLVY-AMC. Means $\pm$ SE, $n = 3$. N.S., $p > 0.05$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.005$ (Student's t -test). Flies were at 14-day-old.

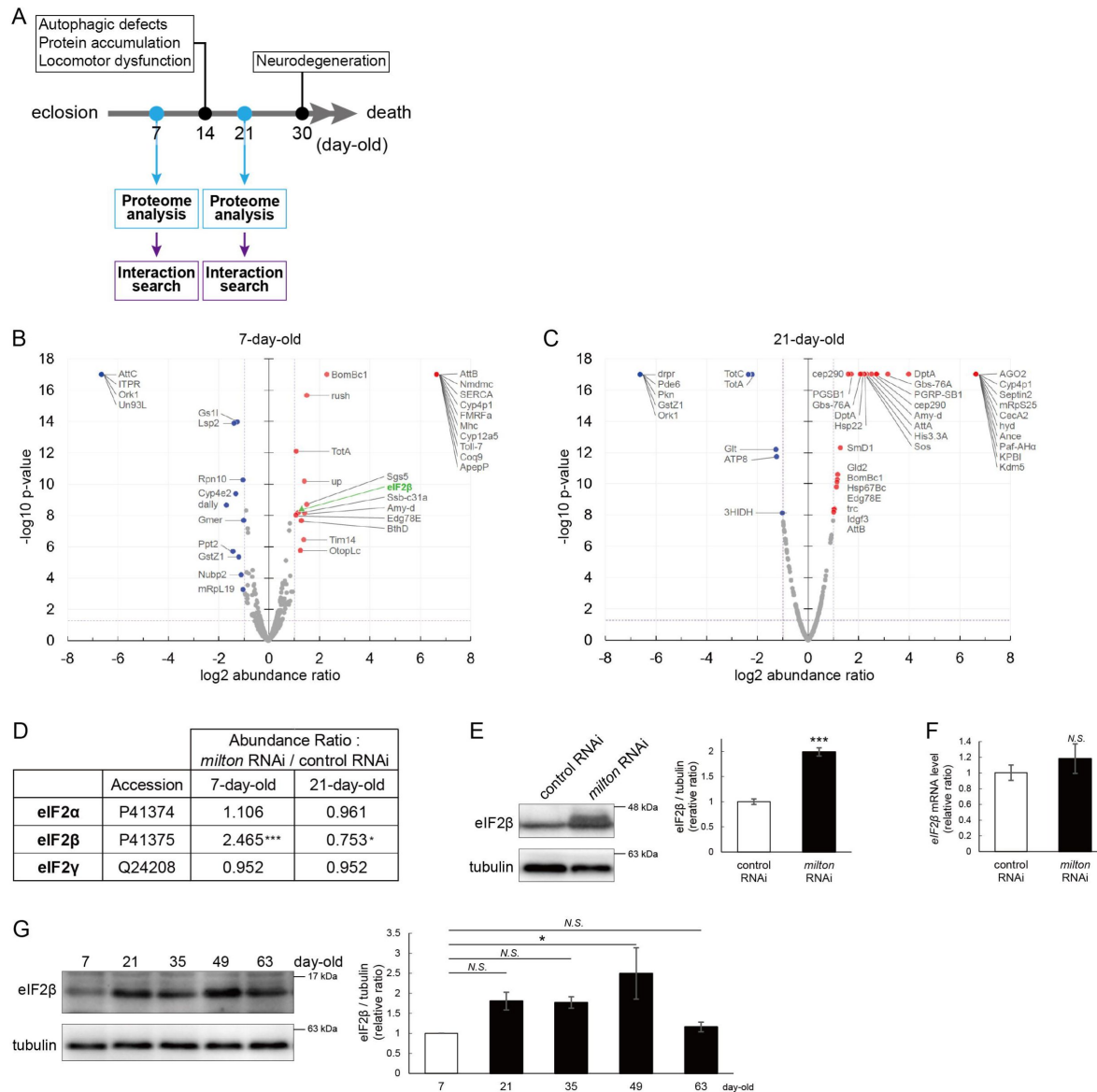


Figure 4.

milton knockdown upregulates eIF2 β in young flies

(A) Timing of proteome analysis and phenotypes observed in *milton* knockdown flies. (B, C) Scatter plots of the $\log_2$ abundance ratio (x-axis) against the $-\log_{10}$ p-value (y-axis) of proteins at 7-day-old (B) and 21-day-old (C). (D) eIF2 subunit protein levels from proteome analysis of *milton* knockdown flies compared to those of control flies. (E) Western blotting of head extracts of flies expressing control or *milton* RNAi in neurons with an anti-eIF2 β antibody. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin was used as a loading control. Means $\pm$ SE, $n = 6$. (F) eIF2 β mRNA levels quantified by qRT-PCR. Means $\pm$ SE, $n = 4$. (G) Western blotting of head extracts of wild-type flies with an anti-eIF2 β antibody. Flies were 7-, 21-, 35-, 49- and 63-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin was used as a loading control. Means $\pm$ SE, $n = 3$, * $p < 0.05$ (one-way analysis of variance (ANOVA) followed by Dunnett's multiple comparison test).

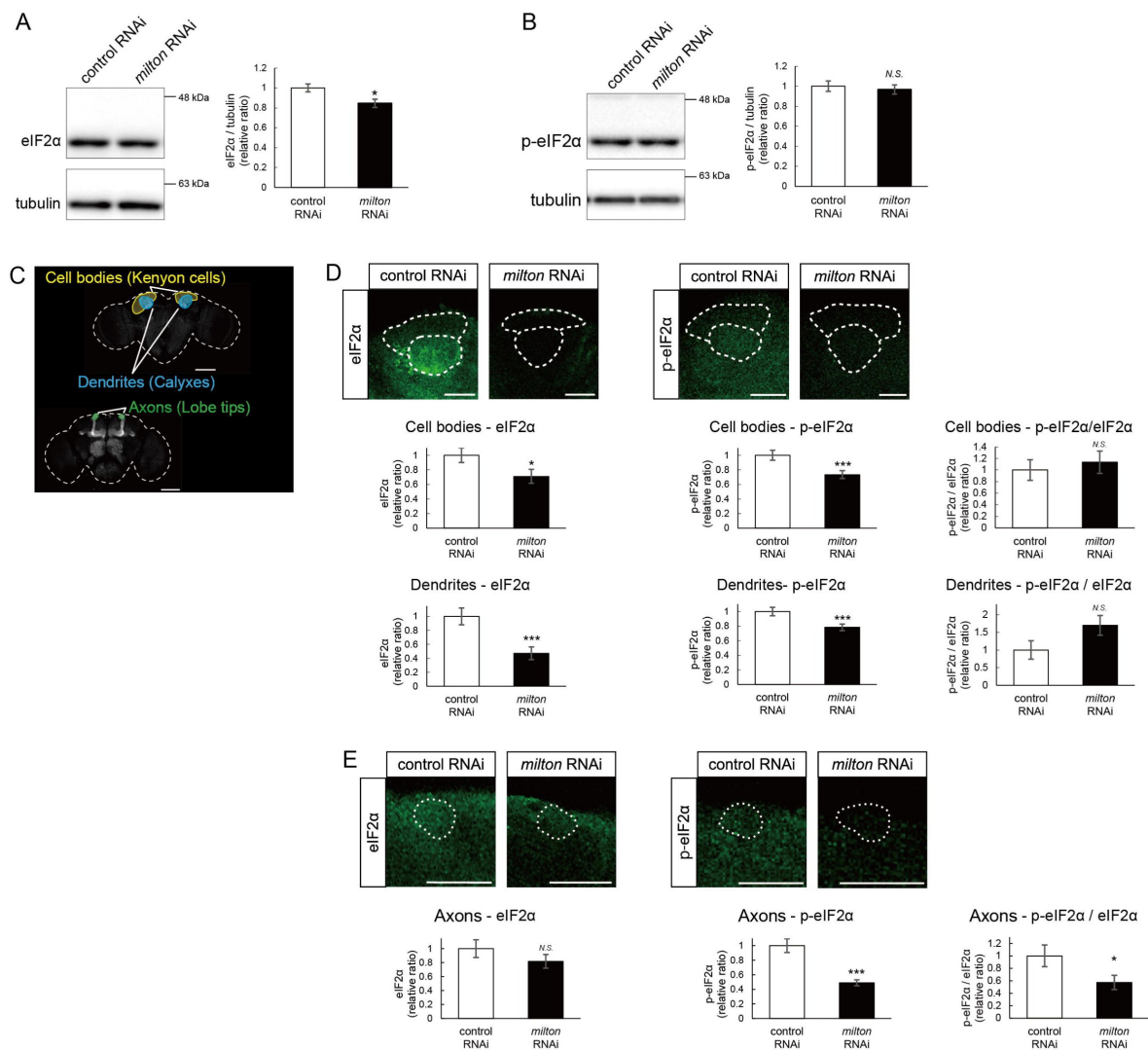


Figure 5.

***milton* knockdown decreases phosphorylation of eIF2α**

(A, B) Western blotting of head extracts with anti-eIF2α (A) and anti-p-eIF2α (B) antibodies. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin was used as a loading control. Means $\pm$ SE, $n = 6$. (C) A schematic representation of the axon (Lobe tips), the cell body region (Kenyon cells), and dendritic region (Calyxes) in the fly brain. Scale bars, 100μm. (D, E) Immunostaining with anti-eIF2α and anti-p-eIF2α antibodies. The mushroom body was identified by expression of mito-GFP. Scale bars, 20μm. The signal intensities of eIF2α and p-eIF2α in axons, dendrites, and cell bodies were quantified and are shown as ratios relative to the control. Means $\pm$ SE, $nC=12$. N.S., $p > 0.05$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.005$ (Student's *t*-test).

To analyze local changes of eIF2 α and p-eIF2 α , we carried out immunostaining. We focused on the mushroom body, where axons, dendrites, and cell bodies can be easily identified (**Figure 5C**). Both eIF2 α and p-eIF2 α were downregulated in the cell body (Kenyon cells) and dendritic (Calyxes) regions of the brains of *milton* knockdown flies (**Figure 5D**). In axons (lobe tips), *milton* knockdown did not affect eIF2 α (**Figure 5E**, $p = 0.271$) but significantly downregulated p-eIF2 α (**Figure 5E**). The ratio of p-eIF2 α to eIF2 α was lower in the axon but not in the soma or dendritic region. These results suggest that axonal distribution of mitochondria regulate the level of overall eIF2 α protein and local p-eIF2 α .

Depletion of axonal mitochondria suppressed global translation

Phosphorylation of eIF2 α induces conformational changes in the eIF2 complex and inhibits global translation³⁶. To analyze the effects of *milton* knockdown on translation, we performed polysome gradient centrifugation to examine the level of ribosome binding to mRNA. Since p-eIF2 α was downregulated, we hypothesized that *milton* knockdown would enhance translation. However, unexpectedly, we found that *milton* knockdown significantly reduced the level of mRNAs associated with polysomes (**Figure 6A and B**). We also compared the level of translation between the brains of control and *milton* knockdown flies by assessing the incorporation of puromycin (**Figure 6C**). Puromycin incorporation was lower in the brains of *milton* knockdown flies than in those of control flies, while it was not statistically significant (**Figure 6C**, indicated by a bracket). These data suggest that the depletion of axonal mitochondria suppresses global translation.

eIF2 β upregulation reduces the level of p-eIF2 α , impairs autophagy, and decreases locomotor function

We were motivated to ask if eIF2 β upregulation mediates autophagic defects caused by *milton* knockdown. If so, neuronal overexpression of *eIF2 β* would also induce autophagy impairment. Neuronal overexpression of *eIF2 β* increased LC3-II, while the LC3-II/LC3-I ratio was not significantly different (**Figure 7A and B**). Overexpression of *eIF2 β* significantly increased the p62 level in the Triton X-100-soluble fraction (**Figure 7C**, 4-fold vs. control, $p < 0.005$ (1% Triton X-100)) but not in the SDS-soluble fraction (**Figure 7C**, 2-fold vs. control, $p = 0.062$ (2% SDS)), as observed in brains of *milton* knockdown flies (**Figure 2B**). These data suggest that neuronal overexpression of *eIF2 β* accumulates autophagic substrates.

Since *milton* knockdown reduced the p-eIF2 α level (**Figure 5E**), we asked whether an increase in eIF2 β affects p-eIF2 α . Neuronal overexpression of *eIF2 β* did not affect the eIF2 α level but significantly decreased the p-eIF2 α level (**Figure 7D and E**).

Depletion of axonal mitochondria causes age-dependent decline in locomotor function²⁴. We found that neuronal overexpression of *eIF2 β* also caused locomotor dysfunction (**Figure 7F**). Locomotor functions were significantly impaired in those flies at 20-day-old and worsened further during aging (**Figure 7F**, compare 4-, 20-, and 30-day-old). We asked if *eIF2 β* overexpression causes neurodegeneration, as depletion of axonal mitochondria in the photoreceptor neurons causes axon degeneration in an age-dependent manner²⁴. *eIF2 β* overexpression in photoreceptor neurons tends to increase neurodegeneration in aged flies, while it was not statistically significant ($p > 0.05$, Figure S2).

These data indicate that an increase of eIF2 β in neurons phenocopies depletion of axonal mitochondria, including suppression of autophagy and age-dependent locomotor dysfunction, and suggest that increase of eIF2 β mediates these phenotypes downstream of loss of axonal mitochondria.

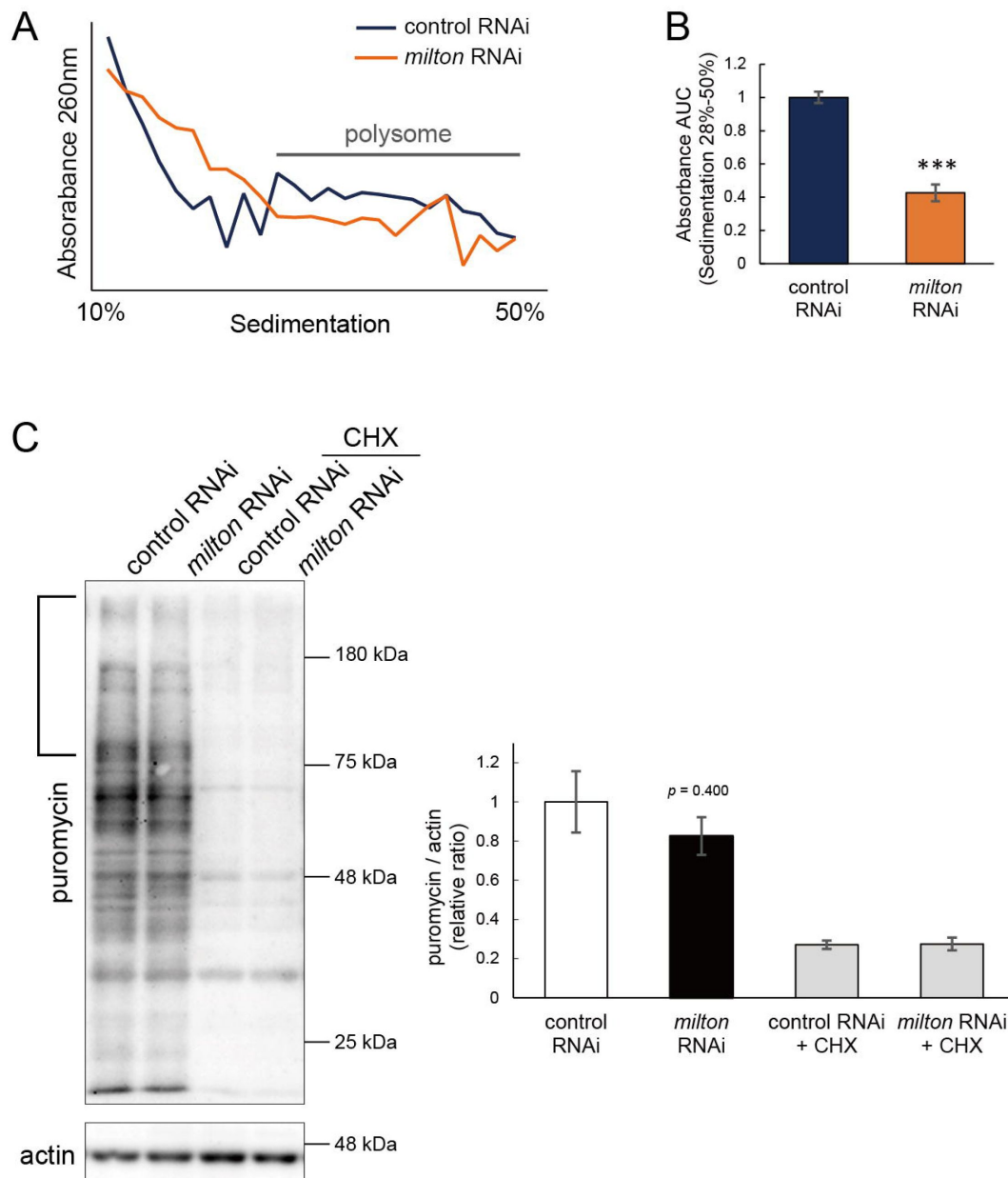


Figure 6.

***milton* knockdown suppressed global translation**

(A) Representative polysome traces of head lysates of control and *milton* knockdown flies. (B) Quantitation of polysome fraction. The relative ratio of area under the curve (AUC) of polysome fractions (sedimentation 28%-50%). Means $\pm$ SE, $n = 3$. *** $p < 0.005$ (Student's *t*-test) (C) Western blotting of head lysates of control and *milton* knockdown flies fed puromycin alone or puromycin and cycloheximide (CHX) with an anti-puromycin antibody. Flies were 14-day-old. Actin was used as a loading control. Representative blots (left) and quantitation (right) are shown. Means $\pm$ SE, $n = 3$. Student's *t*-test.

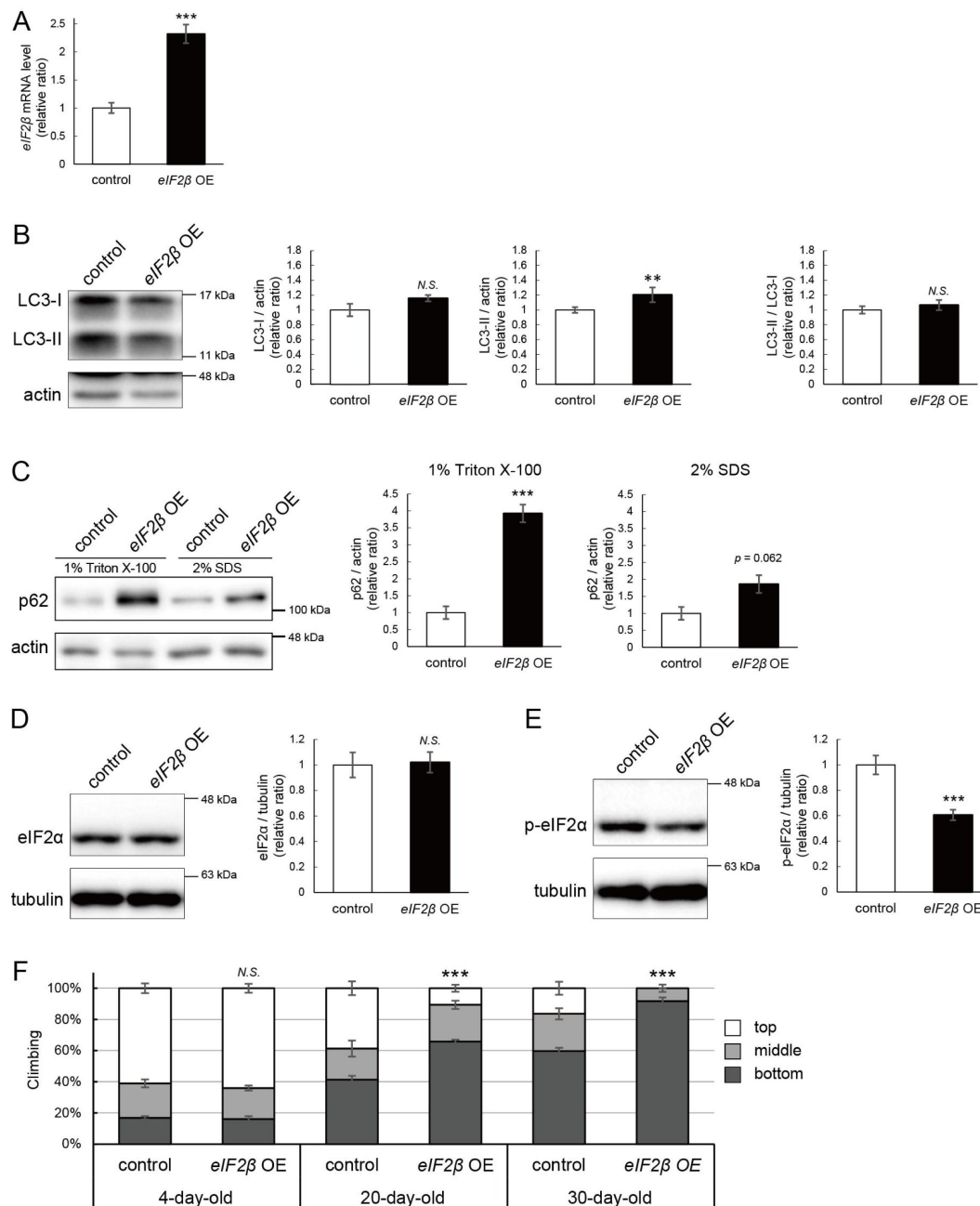


Figure 7.

eIF2β upregulation impairs autophagy and decreases locomotor function

(A) *eIF2β* mRNA levels in head extracts of flies with UAS-*eIF2β* driven by elav-Gal4 (*eIF2β* OE) or UAS-GFP driven by elav-Gal4 (control) were quantified by qRT-PCR. Flies were 2-day-old. Means $\pm$ SE, n=4. (B, C) Western blotting of head extracts with anti-LC3 (B) and anti-p62 (C) antibodies. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin and actin were used as loading controls. Means $\pm$ SE, n = 3 (p62), n = 5 (LC3). (D, E) Western blotting of head extracts with anti-eIF2α (D) and anti-p-eIF2α (E) antibodies. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin was used as a loading control. Means $\pm$ SE, n = 6. (F) Climbing assay revealed early-onset of age-dependent locomotor defects in *eIF2β*-overexpressing flies. Means $\pm$ SE, n = 5. N.S., $p > 0.05$; *** $p < 0.005$ (Student's *t*-test).

Lowering *eIF2 β* rescues autophagic impairment and locomotor dysfunction induced by *milton* knockdown

Finally, we investigated whether suppression of *eIF2 β* rescues autophagy impairment and locomotor dysfunction caused by neuronal knockdown of *milton*. Null mutants and flies with RNAi-mediated knockdown of *eIF2 β* in neurons did not survive. Flies lacking one copy of the *eIF2 β* gene survived without any gross abnormality, and the level of *eIF2 β* mRNA in these flies was about 80% of that in control flies (**Figure 8A**). *eIF2 β* heterozygosity did not affect the *eIF2 α* and p-*eIF2 α* levels (**Figure S3A and B**).

Neuronal knockdown of *milton* causes accumulation of autophagic substrate p62 in the Triton X-100-soluble fraction (**Figure 2B**), and we tested if lowering *eIF2 β* ameliorates it. We found that *eIF2 β* heterozygosity caused a mild increase in LC3-I levels and decreases in LC3-II levels, resulting in a significantly lower LC3-II/LC3-I ratio in *milton* knockdown flies (**Figure 8B**). *eIF2 β* heterozygosity decreased the p62 level in the Triton X-100-soluble fraction in the brains of *milton* knockdown flies (**Figure 8C**). The p62 level in the SDS-soluble fraction, which is not sensitive to *milton* knockdown (**Figure 2B**), was not affected (**Figure 8C**). These results suggest that suppression of *eIF2 β* ameliorates the impairment of autophagy caused by *milton* knockdown.

eIF2 β heterozygosity also rescued locomotor dysfunction induced by *milton* knockdown. *milton* knockdown flies with *eIF2 β* heterozygosity exhibited better locomotor function than *milton* knockdown alone (**Figure 8D**). The *milton* mRNA level was not increased in these flies, indicating that the rescue effect in the *eIF2 β* heterozygous background was not mediated by an increase in the *milton* mRNA level (**Figure S3C**). These data suggest that *eIF2 β* upregulation mediates autophagy impairment and locomotor dysfunction caused by the depletion of axonal mitochondria.

Discussion

The depletion of axonal mitochondria and accumulation of abnormal proteins are both characteristics of aged brains^{37,38}. Proteostasis perturbations trigger the formation of pathological aggregates and increase the risks of neurodegenerative diseases during aging. By using neuronal *milton* knockdown to deplete mitochondria from the axon, we provide evidence that loss of axonal mitochondria drives age-related proteostasis collapse via *eIF2 β* (**Figure 9**). We observed declines in autophagy-mediated degradation of less-aggregated proteins and proteasome activity in *milton* knockdown flies (**Figure 2**). Accumulation of ubiquitinated proteins and changes in age-related pathways started prematurely in *milton* knockdown flies (**Figure 1** and Table 2). *milton* knockdown increased *eIF2 β* and lowered *eIF2 α* phosphorylation in young fly brain (**Figure 4** and **5**). Overexpression of *eIF2 β* phenocopied the effects of *milton* knockdown, including reduced autophagy and accelerated age-related locomotor defects (**Figure 7**). Furthermore, lowering *eIF2 β* levels suppressed the impairment of autophagy and locomotor dysfunction induced by *milton* knockdown (**Figure 8**). From these results, we propose that upregulation of *eIF2 β* downstream of depletion of axonal mitochondria drives age-dependent collapse of proteostasis (**Figure 9**). Our results suggest that mitochondrial distribution and *eIF2 β* are part of the mechanisms constituting proteostasis.

milton knockdown causes loss of mitochondria in the axon and accumulation of mitochondria in the soma. Thus, the detrimental effects may be mediated by the accumulation of mitochondria. However, degeneration induced by *milton* knockdown is prominent in the axon and not detected in the cell body²⁴. Furthermore, abnormal protein accumulation was observed in the axon (**Figure 1**), and p-*eIF2 α* /*eIF2 α* was decreased in the neurites but not in the soma (**Figure 5**), suggesting that proteostasis defects studied in this work are caused by depletion of mitochondria

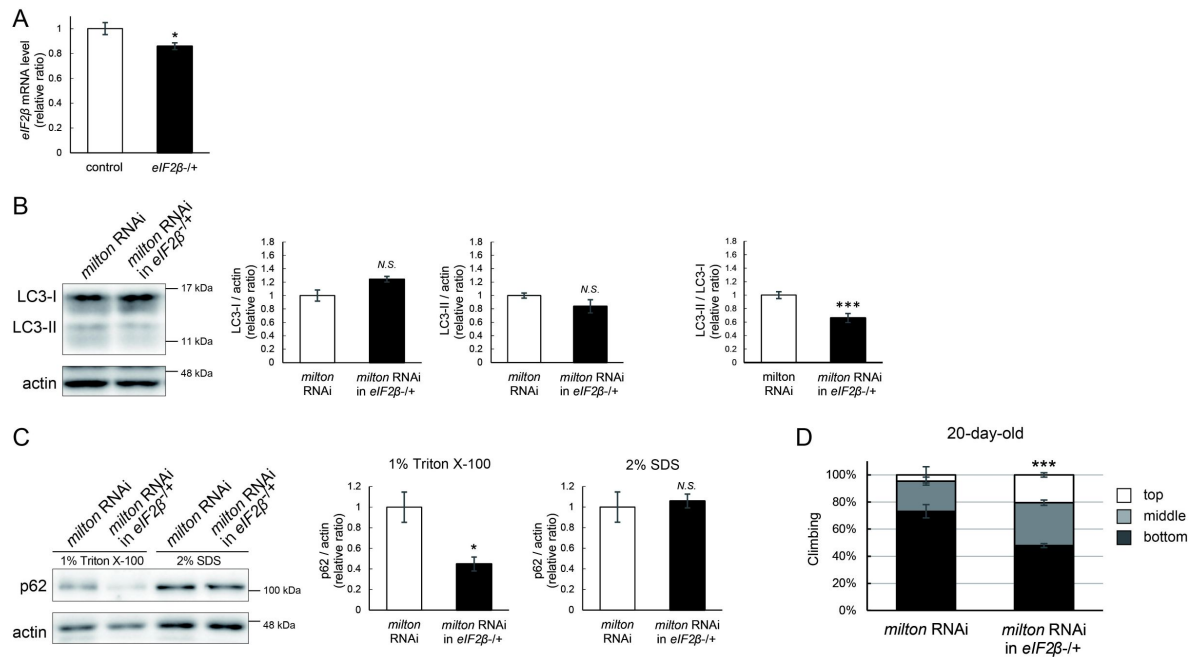


Figure 8.

Lowering *eIF2β* rescues autophagic impairment and locomotor dysfunction induced by *milton* knockdown

(A) *eIF2β* mRNA levels with one disrupted copy of the *eIF2β* gene (*eIF2β*^{SAsstopDsRed/+} (*eIF2β*^{-/+})). Head extracts of flies 2-3 day-old were analyzed by qRT-PCR. Means $\pm$ SE, $n = 3$. (B, C) Western blotting of head extracts of flies with neuronal expression of *milton* RNAi with or without *eIF2β* heterozygosity with anti-LC3 (B) and anti-p62 (C) antibodies. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Actin was used as a loading control. Means $\pm$ SE, $n = 5$ (LC3), $n = 3$ (p62). (D) The climbing ability of 20-day-old flies expressing *milton* RNAi with or without *eIF2β* heterozygosity. Means $\pm$ SE, $n = 15$. N.S., $p > 0.05$; * $p < 0.05$; *** $p < 0.005$ (Student's *t*-test).

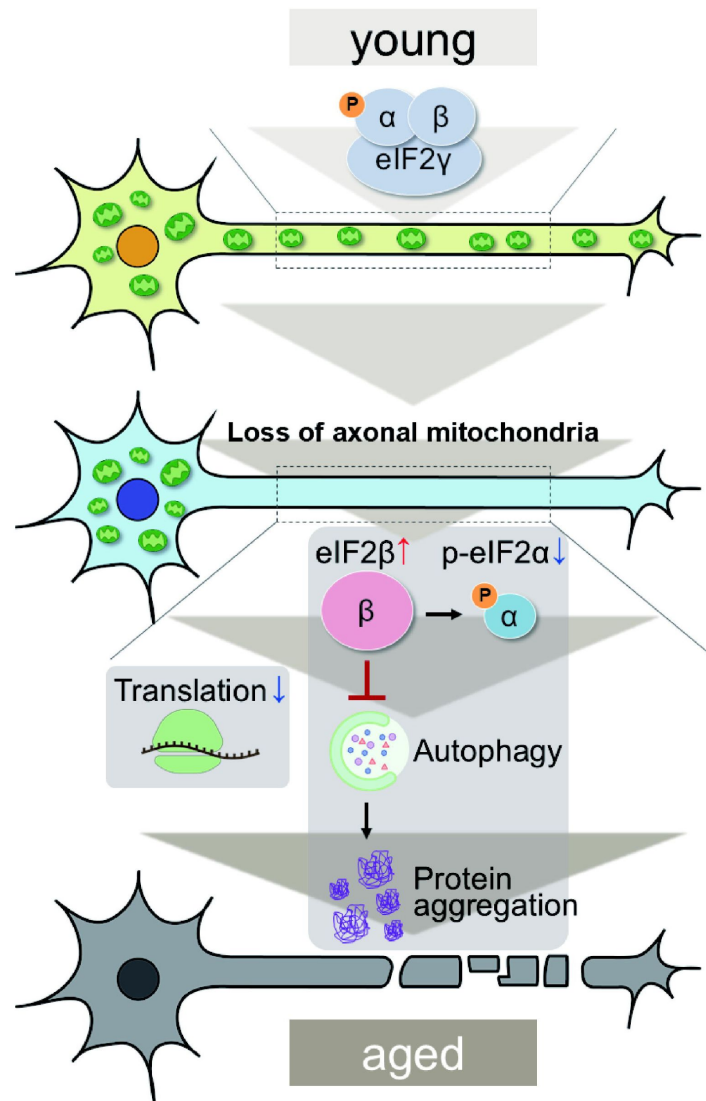


Figure 9.

Graphical abstract: the mitochondria-eIF2 β axis in the axon maintains neuronal proteostasis during aging

rather than accumulation of mitochondria. Further analyses to dissect the effects of *milton* knockdown on proteostasis and translation in the cell body and axon by experiments with spatial resolution would be needed.

Our results suggest that the loss of axonal mitochondria is an event upstream of proteostasis collapse during aging. The number of puncta of ubiquitinated proteins was higher in *milton* knockdown at 14-day-old, but there was no significant difference at 30-day-old (**Figure 1**). Proteome analyses also showed that age-related pathways, such as immune responses, are enhanced in young flies with *milton* knockdown (Table 2). We also found that eIF2 β protein levels increase in an age-dependent manner until 49-day-old and reduces after that (**Figure 4G**). In the brains with neuronal knockdown of *milton*, eIF2 β levels were higher at 7-day-old than those in control and lower at the 21-day-old (**Figure 4D** and Supplementary table). These results suggest that *milton* knockdown is likely accelerating age-dependent changes rather than increasing their magnitude. Disruption of proteostasis is expected to contribute neurodegeneration³⁸, and it would be interesting to analyze the sequence of protein accumulation and axonal degeneration in *milton* knockdown (^{24,29} and **Figure 1**) in detail with higher time resolution.

Our results revealed that eIF2 β regulates autophagy and maintains proteostasis during aging. eIF2 β is a component of eIF2, which mediates translational regulation and ISR initiation. When ISR is activated, phosphorylated eIF2 α suppresses global translation and induces translation of ATF4, which mediates transcription of autophagy-related genes^{39,40}. Since ISR can positively regulate autophagy, we suspected that suppression of ISR underlies a reduction in autophagic protein degradation. We found neuronal knockdown of *milton* reduced phosphorylated eIF2 α , suggesting that ISR is reduced (**Figure 5**). However, we also found that global translation was reduced (**Figure 6**). Increased levels of eIF2 β might disrupt the eIF2 complex or alter its functions. The stoichiometric mismatch caused by an imbalance of eIF2 components may inhibit ISR induction. Supporting this model, we found that eIF2 β upregulation reduced the levels of p-eIF2 α (**Figure 7**). It is also possible that eIF2 β mediates autophagy defects via mechanisms independent of ISR since eIF2 β has functions independent of eIF2 α ^{41,42}. For example, suppression of *eIF2 β* has been reported to slow down cancer cell growth⁴¹. In developing neurons, eIF2 β can directly interact with the translational repressor Kra to regulate midline axon guidance⁴². Our results also suggest that *milton* knockdown and overexpression of eIF2 β affect autophagy via increased LC3-I abundance (**Figures 2** and **7**), suggesting an unconventional mechanism of autophagy suppression. To our knowledge, the roles of eIF2 β in aging and autophagy independent of ISR have not been reported. Our results revealed a novel function of eIF2 β to maintain proteostasis during aging, while further investigation is required to elucidate underlying mechanisms.

How depletion of axonal mitochondria upregulates eIF2 β is currently under investigation. A major mitochondrial function is ATP production, and depletion of axonal mitochondria downregulates ATP in axons³³. However, we found that ATP deprivation did not always suppress autophagy (**Figure 3**), suggesting it is unlikely to be involved in the mechanisms that induce eIF2 β upregulation. Mitochondria also serve as signaling hubs for translation and protein degradation. Mitochondrial proteins are regulated by co-translational protein quality control, and mitochondrial damage induces translational stalling of mitochondrial outer membrane-associated *complex-I 30 kD subunit (C-I30)* mRNA⁴³. Additionally, the mitochondrial outer membrane ubiquitin ligase MITOL (also known as MARCHF5) ubiquitinates and regulates not only mitochondrial proteins such as Mfn2⁴⁴ but also microtubule-associated⁴⁵ and endoplasmic reticulum⁴⁶ proteins. These findings indicate that mitochondria serve as local signaling centers for proteostasis maintenance, and eIF2 β levels may also be regulated by mechanisms related to mitochondria.

In conclusion, our results suggest that axonal mitochondria and eIF2 β form an axis to maintain constitutive autophagy. Suppression of *eIF2 β* rescued autophagic defects and neuronal dysfunction upon loss of axonal mitochondria. Since eIF2 β is conserved across many species, including *Drosophila* and humans, our results suggest that eIF2 β may be a possible therapeutic target for aging and diseases associated with mitochondrial mislocalization.

Materials and methods

Fly stocks and husbandry

Flies were maintained in standard cornmeal medium (10% glucose, 0.7% agar, 9% cornmeal, 4% yeast extract, 0.3% propionic acid, and 0.1% n-butyl p-hydroxybenzoate) at 25°C under light–dark cycles of 12:12 h. The flies were transferred to fresh food vials for every 2–3 days. UAS-*milton* RNAi (v41508) was from VDRC and outcrossed to [w1118] for five generations in our laboratory. Transgenic fly lines carrying UAS-*luciferase* RNAi control for *milton* RNAi) was reported previously²⁴, GMR-gal4, Elav-gal4, UAS-*Pfk* RNAi (Bloomington stock center #36782), UAS-*luciferase* RNAi (Bloomington stock center #31603) (control for *Pfk* RNAi), UAS-*GFP* (used for control for UAS-*eIF2 β*) and UAS-*eIF2 β* (eIF2 β ^{EY08063}, Bloomington stock center #17425) were from the Bloomington stock center. *eIF2 β* heterozygous strain (PBac{SAsstopDsRed} LL07719, DGRC#142114) was from KYOTO *Drosophila* Stock Center. UAS-*mitoGFP* was a kind gift from Drs. W. M. Saxton (University of California, Santa Cruz).

Immunohistochemistry and image acquisition

Fly brains were dissected in PBS and fixed for 45 minutes in formaldehyde (4% v/v in PBS) at room temperature. After incubation in PBST containing 0.1% Triton X-100 for 10 min three times, samples were incubated for 1h at room temperature in PBST containing 1% normal goat serum (Wako, #143-06561) and then incubated overnight with the primary antibody (anti-ubiquitin antibody Ubi-1 (Thermo Fisher #13-1600) (1:50), anti-eIF2 α (abcam #ab26197) (1:50) and anti-p-eIF2 α (Cell signaling #3398S) (1:50)) diluted in 1% NGS/PBST at 4 °C. Samples were then washed for 10 min with PBST including 0.1% Triton X-100 three times and incubated with the secondary antibody for overnight at 4° C. Brains were mounted in Vectashield (Vectorlab Cat#H-1100) and analyzed under a confocal microscope (Nikon). Quantitative analysis was performed using ImageJ (National Institutes of Health) with maximum projection images derived from Z-stack images acquired with same settings. Puncta were identified with mean intensity and area using ImageJ. For eIF2 α and p- eIF2 α immunostaining, the mushroom body was detected by mitoGFP expression.

Electron microscopy

Proboscis was removed from decapitated heads, which were then incubated in primary fixative solution (2.5% glutaraldehyde and 2% paraformaldehyde in 0.1 M sodium cacodylate buffer) at R.T. for 2 hours. After washing heads with 3% sucrose in 0.1 M sodium cacodylate buffer, fly heads were post-fixed for 1 hour in secondary fixation (1% osmium tetroxide in 0.1 M sodium cacodylate buffer) on ice. After washing with H₂O, heads were dehydrated with ethanol and infiltrated with propylene oxide and Epon mixture (TAAB and Nissin EM) for 3 hours. After infiltration, specimens were embedded with an Epon mixture at 70°C for 2-3 days. Thin sections (70 nm) of laminae were collected on copper grids. The sections were stained with 5% uranyl acetate in 50% ethanol and Reynolds' lead citrate solution. Electron micrographs were obtained with a CCD Camera mounted on a JEM-1400 plus electron microscope (Jeol Ltd.). Quantitation was performed using ImageJ (National Institutes of Health).

SDS-PAGE and immunoblotting

Western blotting was performed as reported previously²⁴. Briefly, heads of 10-20 *Drosophila* were homogenized with SDS-Tris-Glycine sample buffer (0.312M Tris, 5% SDS, 8% glycerol, 0.0625% BPB, 10% β -mercaptoethanol, 10 μ g/mL leupeptin, 0.4 μ M Pefabloc, 10mM β -glycerphosphate, 10mM NaF) and after boiling at 95°C for 2 minutes, it was centrifuged at 13,200 rpm, and the supernatant was used as a sample. For p62 western blot, fly heads were homogenized with 1% PBST and after centrifugation at 13,200 rpm, the supernatant was mixed 1:1 SDS-Tris-Glycine sample buffer, and boiled at 95 °C for 2 minutes. The pellet was dissolved with 2% SDS in PBS, then centrifuged again at 13,200 rpm. The supernatant was mixed 1:1 SDS-Tris-Glycine sample buffer and then boiled at 95 °C for 2 minutes. SDS-PAGE for western blotting was performed using 15%(w/v) (LC3), 7.5%(w/v) (p62), 10% (w/v) (eIF2 α , β , and p-eIF2 α) polyacrylamide gels. After electrophoresis, they were transferred to PVDF membrane (Merck Millipore) using a transfer device (BIO-RAD). After transfer, the membrane was blocked with 5% skim milk/TBST (50 mM Tris (pH 7.5), 0.15 M NaCl, 0.05% Tween20) for 1 hour and incubated with primary antibody listed below overnight at 4 °C. Membranes were rinsed twice with TBST containing 0.65M NaCl and once with TBST containing 0.15M NaCl. After incubation with the secondary antibody at room temperature for 1 hour, membranes were rinsed twice with TBST containing 0.65M NaCl and once with TBST containing 0.15M NaCl. After incubation with Immobilon Western Chemiluminescent HRP Substrate (Merck Millipore), chemiluminescent signals were detected with Fusion FX (Vilber). Experiments were repeated at least 3 times with independent cohorts of flies.

Primary antibodies: anti-LC3 antibody Atg8 (Merck Millipore #ABC975) (1:1000), anti-p62 antibody Ref2P (abcam #ab178440) (1:750), anti-eIF2 β antibody (1:1500), anti-eIF2 α antibody (abcam #ab26197) (1:1000), anti-p-eIF2 α antibody (Cell signaling #3398S) (1:2000), anti-actin antibody (SIGMA A2066) (1:3000) and anti- β tubulin antibody (Sigma #T9026) (1:800,000). Polyclonal anti-eIF2 β antibody was raised against a synthetic peptide (CGLEDDTKKEDPQDEA) corresponding to the C-terminal residues 29-43 of *Drosophila* eIF2 β (1:1500).

Secondary antibodies: Peroxidase-conjugated goat anti-mouse IgG antibody (Dako #P0447) (1:2000), peroxidase-conjugated pig anti-rabbit IgG antibody (Dako #P0399) (1:2000)

Proteasome assay

Heads from ten flies were homogenized in 150 μ l of buffer B (25 mM Tris-HCl [pH 7.5], 2 mM ATP, 5 mM MgCl₂, and 1 mM dithiothreitol). Proteasome peptidase activity in the lysates was measured with a synthetic peptide substrate, succinyl-Leu-Leu-Val-Tyr-7-amino-4-methyl-coumarin (Suc-LLVY-AMC) (Cayman). Luminescence was measured on a multimode plate reader 2300 Enspire (PerkinElmer). Experiments were repeated at least 3 times with independent cohorts of flies.

ATP assay

Heads from the 10 flies were homogenized in 50 μ l of 6CM guanidine-HCl in extraction buffer (100mM Tris and 4mM EDTA, pH 7.8) to inhibit ATPases. Samples were boiled for 5Cmin and centrifuged. The supernatant was diluted 4% with extraction buffer and mixed with a reaction solution (ATP Determination kit, Invitrogen). Luminescence was measured on a multimode plate reader 2300 Enspire (PerkinElmer). The relative ATP levels were calculated by dividing the luminescence by the total protein concentration, which was determined by the Bradford method. Experiments were repeated at least 3 times with independent cohorts of flies.

Proteomic assay and pathway analysis

Sample preparation

Heads from the 35 flies were homogenized in 110 μ l of extraction buffer (0.25% RapiGest SF, 50mM ammonium bicarbonate, 10mM dithiothreitol, 10 μ g/mL leupeptin, 0.4 μ M Pefabloc, 10mM β -glycerophosphate, 10mM NaF). Homogenized samples were centrifuged and boiled for 5Cmin. After quantification of the protein concentration using a Pierce® 660 nm Protein Assay (Thermo Fisher Scientific), 10 μ g proteins from each sample were reduced using 5 mM tris (2-carboxyethyl) phosphine hydrochloride (TCEP-HCl; Thermo Fisher Scientific) at 60°C for 1 h, alkylated using 15 mM iodoacetamide (Fujifilm Wako Pure Chemical, Osaka, Japan) at room temperature for 30 min, and then digested using 1.5 μ g Trypsin Gold (Mass Spectrometry Grade; Promega, Madison, WI, USA) at 37°C for 17 h. The digests were acidified by the addition of trifluoroacetic acid (TFA), incubated at 37°C for 30 min, and then centrifuged at 17,000 $\times$ g for 10 min to remove the RapiGest SF. The supernatants were collected and desalted using MonoSpin™ C18 (GL Sciences, Tokyo, Japan). The resulting eluates were concentrated *in vacuo*, dissolved in 2% MeCN containing 0.1% formic acid (FA), and subjected to LC-MS/MS analysis.

LC-MS/MS analysis and database search

LC-MS/MS analyses were performed on an Ultimate 3000 RSLCnano system (Thermo Fisher Scientific) coupled to a Q Exactive hybrid quadrupole-Orbitrap mass spectrometer (Thermo Fisher Scientific) equipped with a nano electron spray ionization (ESI) source. The LC system was equipped with a trap column (C18 PepMap 100, 0.3 $\times$ 5 mm, 5 μ m, Thermo Fisher Scientific) and an analytical column (NTCC-360/75-3-125, Nikkyo Technos, Tokyo, Japan). Peptide separation was performed using a 90-min gradient of water/0.1% FA (mobile phase A) and MeCN/0.1% FA (mobile phase B) at a flow rate of 300 nL/min. Elution was performed as follows: 0–3 min, 2% B; 3–93 min, 2%–40% B; 93–95 min, 40%–95% B; 95–105 min, 95% B; 105–107 min, 95%–2% B; and 107–120 min, 2% B. The mass spectrometer was operated in data-dependent acquisition mode. The MS parameters were as follows: spray voltage, 2.0 kV; capillary temperature, 275°C; S-lens RF level, 50; scan type, full MS; scan range, m/z 350–1500; resolution, 70,000; polarity, positive; automatic gain control target, 3×10^6 ; and maximum injection time, 100 msec. The MS/MS parameters were as follows: resolution, 17,500; automatic gain control target, 1×10^5 ; maximum injection time, 60 msec; normalized collision energy (NCE), 27; dynamic exclusion, 15 sec; loop count, 10; isolation window, 1.6 m/z ; charge exclusion: unassigned, 1 and ≥ 8 ; and injection volume, 1 μ L (containing 0.5 μ g protein). Measurements were made in duplicate for each sample.

The identification of proteins and label-free quantification (LFQ) of the detected peptides were performed using Proteome Discoverer software ver. 2.4 (Thermo Fisher Scientific). The analytical parameters used for the database search were as follows: parent mass error tolerance, 10.0 ppm; fragment mass error tolerance, 0.02 Da; search engine, sequest HT; protein database, *Drosophila melanogaster* (Fruit fly: SwissProt Tax ID=7227); enzyme name, trypsin (full); maximum number of missed cleavages, 2; dynamic modification, oxidation (methionine), phosphorylation (serine, threonine, tyrosine), acetyl (lysine), GG (lysine); N-terminal modification, Met-loss (methionine), and Met-loss+acetyl (methionine); static modification, carbamidomethylation (cysteine) and FDR confidence, High < 0.01, 0.01 $\leq$ Medium < 0.05, 0.05 $\leq$ Low. The parameters for LFQ were as follows: precursor abundance, based on area; and normalization mode, total peptide amount.

The abundance ratio of *milton* RNAi to control RNAi at 7- or 21-day-old was calculated. We considered proteins with an abundance ratio of ≥ 2.0 or ≤ 0.5 and an ANOVA *P*-value of < 0.05 based on volcano plots to be differentially expressed of *milton* RNAi. To extract molecular networks biologically relevant to the proteins that are differentially expressed in *milton* RNAi, pathway analysis was performed using KeyMolnet (KM Data Inc., Tokyo, Japan).

RNA extraction and quantitative real-time PCR analysis

Heads from more than 25 flies were mechanically isolated, and total RNA was extracted using ISOGEN (NipponGene) followed by reverse-transcription using PrimeScript RT reagent kit (Takara). The resulting cDNA was used as a template for PCR with THUNDERBIRD SYBR qPCR mix (TOYOBO) on a Thermal Cycler Dice real-time system TP800 (Takara). Expression of genes of interest was standardized relative to *rp49*. Relative expression values were determined by the $\Delta\Delta CT$ method (Livak and Schmittgen, 2001). Experiments were repeated three times, and a representative result was shown.

Primers were designed using DRSC FlyPrimerBank (Harvard Medical School). Primer sequences are shown below;

eIF2 β for 5'-GGACGACGACAAGAGCGAAG -3'

eIF2 β rev 5'-CGGTCGCATCACGAACCTTG -3'

milton for 5'-GGCTTCAGGGCCAGGTATCT-3'

milton rev 5'-GCCGAACCTGGCTGACTTTG-3'

Actin for 5'-TGCACCGCAAGTGCTTCTAA-3'

Actin rev 5'-TGCTGCACTCCAAACTTCCA-3'

rp49 for 5'-GCTAAGCTGTCGCACAAATG-3'

rp49 rev 5'- GTTCGATCCGTAACCGATGT-3'

Polysome gradient centrifugation

30 heads were homogenized in 150 μ l of lysis buffer (25 mM Tris pH 7.5, 50 mM MgCl₂, 250 mM NaCl, 1 mM DTT, 0.5 mg/ml cycloheximide, 0.1 mg/ml heparin). The lysates were centrifuged at 13,200 rpm at 4 °C for 5 minutes, and the supernatant was collected. The samples containing 38 μ g of RNA was layered gently on top of a 10–50% w/w sucrose gradient (50 mM Tris pH 7.5, 50 mM MgCl₂, 250 mM NaCl, 0.1 mg/ml heparin, 0.5 mg/ml cycloheximide in 5 ml polyallomer tube) and centrifuged at 37,000 rpm at 4°C for 150 minutes in a himac CP-NX ultracentrifuge using a P50AT rotor. Samples were fractionated from top to bottom, and absorbance at OD260 nm was analyzed by a Plate reader (EnSpire). Experiments were repeated at least 3 times with independent cohorts of flies.

Puromycin analysis

13-day-old flies were starved for 6 hr and fed 600 μ M puromycin (Sigma) or 600 μ M puromycin/35 mM cycloheximide (Sigma) in 5% sucrose solution for 20 hr. Incorporated puromycin was quantified by western blot with anti-puromycin antibody and normalized with actin. Experiments were repeated at least 3 times with independent cohorts of flies.

Histological analysis

Fly heads were fixed in Bouin's fixative solution for 48 h at room temperature, incubated for 24 h in 50 mM Tris/150 mM NaCl, and embedded in paraffin. Serial sections (7 μ m thickness) through the entire heads were stained with hematoxylin and eosin and examined by bright-field

microscopy. Images of the sections that include the lamina were captured with Keyence microscope BZ-X700 (Keyence), and the vacuole area was measured using ImageJ (National Institutes of Health).

Climbing assay

The climbing assay was performed as previously described²⁴[\[1\]](#). Flies were placed in an empty plastic vial (2.5 cm in diameter × 10 cm in length). The vial was gently tapped to knock the flies to the bottom, and the number of flies reached to the top, middle and bottom areas of the vials in 10 s was counted. Experiments were repeated 10 times, and the mean percentage of flies in each area and standard deviations were calculated. Experiments were repeated with independent cohorts more than three times, and a representative result was shown.

Statistics

The number of replicates, what n represents, precision measurements, and the meaning of error bars are indicated in Figure Legends. Data are shown as means ± SEM. For pairwise comparisons, Student's t-test was performed with Microsoft Excel (Microsoft). For multiple comparisons, data were analyzed using one-way ANOVA with Tukey's HSD multiple-comparisons test in the GraphPad Prism 6.0 software (GraphPad Software, Inc., La Jolla, CA). Results with a p-value of less than 0.05 were considered to be statistically significant.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon request.

Acknowledgements

The authors thank the Bloomington stock center; TRiP at Harvard Medical School (NIH/NIGMS R01-GM084947); the Kyoto Drosophila Stock Center and the Vienna Drosophila RNAi Center for fly stocks. The authors thank Dr. Masayuki Miura from Department of Pharmaceutical Science, University of Tokyo, for proteasome activity assay protocol; Dr. Shin-ichi Hisanaga from the Department of Biological Sciences, Tokyo Metropolitan University, for critical comments; Dr. Taro Saito, Dr. Akiko Asada from the Department of Biological Sciences, Tokyo Metropolitan University, Dr. Michiko Sekiya from Department of Alzheimer's Disease Research, National Center for Geriatrics and Gerontology, and Dr. Seiji Watanabe, Dr. Koji Yamanaka from Department of Neuroscience and Pathobiology, Research Institute of Environmental Medicine, Nagoya University for technical supports.

Additional information

Funding

This work was supported by the Sasakawa Scientific Research Grant (2021-4087) (to K.S.), the Takeda Science Foundation (to KA), Hoansha foundation grant (to KA), a research award from the Japan Foundation for Aging and Health (to KA), the Novartis Foundation (Japan) for the promotion of Science (to KA), JSPS KAKENHI Grant-in-Aid for Scientific Research on Challenging Research (Exploratory) JP19K21593 (to KA), JSPS KAKENHI Grant-in-Aid for Scientific Research(B) JP24K02860 (to KA), NIG-JOINT (National Institute of Genetics, 71A2018, 25A2019) (to K.A.) and TMU strategic research fund for social engagement (to KA).

Author contributions

K.S. and K.A. designed the experiments, interpreted the data, and wrote the manuscript; K.S., Y.M., K.M.I and E.S. conducted the experiments, and K.S., Y.M., E.S., and K.A. analyzed the data. All authors read and approved the manuscript.

Funding

Sasakawa Scientific Research Grant (2021-4087)

Yoga Science Foundation

Hoansha foundation grant

Japan Foundation for Aging and Health

Novartis Foundation (Japan) for the promotion of Science

Japan Society for the Promotion of Science (JP19K21593)

Japan Society for the Promotion of Science (JP24K02860)

National Institute of Genetics (NIG-JOINT 71A2018)

National Institute of Genetics (NIG-JOINT 25A2019)

Additional files

Table S1, Table S2 [↗](#)

Figure S1. Ubiquitinated proteins in brains with neuronal knockdown of *milton* at 1-day-old. Brains dissected at 1-day-old were immunostained with an antibody against ubiquitin. Representative images (left) and quantitation of the number of ubiquitin-positive puncta (right) are shown. Scale bars of hemibrains, 100 μ m, Means $\pm$ SE, $n = 8$. N.S., $p > 0.05$ (Student's *t*-test). [↗](#)

Figure S2. Histology analysis of fly heads with *eIF2 β* overexpression. The morphology of the eye with *eIF2 β* overexpression. The dotted lines indicate the retina. Vacuoles are indicated by arrows. Representative images (left) and quantification of vacuole area (right). The flies were 40-day-old. Means $\pm$ SE, $n = 5-13$ N.S., $p > 0.05$ (Student's *t*-test). [↗](#)

Figure S3. Lowering the *eIF2 β* level does not affect the levels of *eIF2 α* and *p-eIF2 α* . (A, B) Blotting was performed with anti-*eIF2 α* (A) and anti-*p-eIF2 α* (B) antibodies. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin was used as a loading control. Means $\pm$ SE, $n = 6$. (C) *eIF2 β* gene disruption does not affect the knockdown efficiency of *milton*. *milton* mRNA levels in head extracts were quantified by qRT-PCR. Flies were 2-day-old. Means $\pm$ SE, $n = 3$. N.S., $p > 0.05$; * $p < 0.05$ (Student's *t*-test). [↗](#)

References

1. Yerbury J.J., Ooi L., Dillin A., Saunders D.N., Hatters D.M., Beart P.M., Cashman N.R., Wilson M.R., Ecroyd H (2016) **Walking the tightrope: proteostasis and neurodegenerative disease** *J.*

Neurochem **137**:489–505 <https://doi.org/10.1111/jnc.13575> | Google Scholar

2. Hetz C (2021) **Adapting the proteostasis capacity to sustain brain healthspan** *Cell* **184**:1545–1560 <https://doi.org/10.1016/j.cell.2021.02.007> | Google Scholar
3. Balch W.E., Morimoto R.I., Dillin A., Kelly J.W (2008) **Adapting proteostasis for disease intervention** *Science* **319**:916–919 <https://doi.org/10.1126/science.1141448> | Google Scholar
4. Kimball S.R (1999) **Eukaryotic initiation factor eIF2** *Int J Biochem Cell Biol* **31**:25–29 [https://doi.org/10.1016/s1357-2725\(98\)00128-9](https://doi.org/10.1016/s1357-2725(98)00128-9) | Google Scholar
5. Jackson R.J., Hellen C.U., Pestova T.V (2010) **The mechanism of eukaryotic translation initiation and principles of its regulation** *Nat. Rev. Mol. Cell Biol* **11**:113–127 <https://doi.org/10.1038/nrm2838> | Google Scholar
6. Pakos-Zebrucka K., Koryga I., Mnich K., Ljubic M., Samali A., Gorman A.M (2016) **The integrated stress response** *EMBO Rep* **17**:1374–1395 <https://doi.org/10.15252/embr.201642195> | Google Scholar
7. Kroemer G., Marino G., Levine B (2010) **Autophagy and the integrated stress response** *Mol. Cell* **40**:280–293 <https://doi.org/10.1016/j.molcel.2010.09.023> | Google Scholar
8. Nandi D., Tahiliani P., Kumar A., Chandu D (2006) **The ubiquitin-proteasome system** *J Biosci* **31**:137–155 <https://doi.org/10.1007/BF02705243> | Google Scholar
9. Glick D., Barth S., Macleod K.F (2010) **Autophagy: cellular and molecular mechanisms** *J. Pathol* **221**:3–12 <https://doi.org/10.1002/path.2697> | Google Scholar
10. Vargas J.N.S., Hamasaki M., Kawabata T., Youle R.J., Yoshimori T (2023) **The mechanisms and roles of selective autophagy in mammals** *Nat. Rev. Mol. Cell Biol* **24**:167–185 <https://doi.org/10.1038/s41580-022-00542-2> | Google Scholar
11. Aman Y., Schmauck-Medina T., Hansen M., Morimoto R.I., Simon A.K., Bjedov I., Palikaras K., Simonsen A., Johansen T., Tavernarakis N., et al. (2021) **Autophagy in healthy aging and disease** *Nat Aging* **1**:634–650 <https://doi.org/10.1038/s43587-021-00098-4> | Google Scholar
12. Ross C.A., Poirier M.A (2004) **Protein aggregation and neurodegenerative disease** *Nat. Med* **10**:S10–17 <https://doi.org/10.1038/nm1066> | Google Scholar
13. Rubinshtein D.C., Marino G., Kroemer G (2011) **Autophagy and aging** *Cell* **146**:682–695 <https://doi.org/10.1016/j.cell.2011.07.030> | Google Scholar
14. Vos M., Lauwers E., Verstreken P (2010) **Synaptic mitochondria in synaptic transmission and organization of vesicle pools in health and disease** *Front Synaptic Neurosci* **2** <https://doi.org/10.3389/fnsyn.2010.00139> | Google Scholar
15. Cheng A., Hou Y., Mattson M.P (2010) **Mitochondria and neuroplasticity** *ASN Neuro* **2**:e00045 <https://doi.org/10.1042/AN20100019> | Google Scholar
16. Hollenbeck P.J., Saxton W.M (2005) **The axonal transport of mitochondria** *J. Cell Sci* **118**:5411–5419 <https://doi.org/10.1242/jcs.02745> | Google Scholar
17. Takihara Y., Inatani M., Eto K., Inoue T., Kreymerman A., Miyake S., Ueno S., Nagaya M., Nakanishi A., Iwao K., et al. (2015) **In vivo imaging of axonal transport of mitochondria in**

the diseased and aged mammalian CNS *Proc Natl Acad Sci U S A* **112**:10515–10520 <https://doi.org/10.1073/pnas.1509879112> | Google Scholar

18. Milde S., Adalbert R., Elaman M.H., Coleman M.P (2015) **Axonal transport declines with age in two distinct phases separated by a period of relative stability** *Neurobiol. Aging* **36**:971–981 <https://doi.org/10.1016/j.neurobiolaging.2014.09.018> | Google Scholar
19. Vagnoni A., Hoffmann P.C., Bullock S.L (2016) **Reducing Lissencephaly-1 levels augments mitochondrial transport and has a protective effect in adult Drosophila neurons** *J. Cell Sci* **129**:178–190 <https://doi.org/10.1242/jcs.179184> | Google Scholar
20. Morsci N.S., Hall D.H., Driscoll M., Sheng Z.H (2016) **Age-Related Phasic Patterns of Mitochondrial Maintenance in Adult Caenorhabditis elegans Neurons** *J. Neurosci* **36**:1373–1385 <https://doi.org/10.1523/JNEUROSCI.2799-15.2016> | Google Scholar
21. Adalbert R., Coleman M.P (2013) **Review: Axon pathology in age-related neurodegenerative disorders** *Neuropathol. Appl. Neurobiol* **39**:90–108 <https://doi.org/10.1111/j.1365-2990.2012.01308.x> | Google Scholar
22. Duncan J.E., Goldstein L.S (2006) **The genetics of axonal transport and axonal transport disorders** *PLoS Genet* **2**:e124 <https://doi.org/10.1371/journal.pgen.0020124> | Google Scholar
23. Chen H., Chan D.C (2009) **Mitochondrial dynamics--fusion, fission, movement, and mitophagy--in neurodegenerative diseases** *Hum. Mol. Genet* **18**:R169–176 <https://doi.org/10.1093/hmg/ddp326> | Google Scholar
24. Iijima-Ando K., Sekiya M., Maruko-Otake A., Ohtake Y., Suzuki E., Lu B., Iijima K.M (2012) **Loss of axonal mitochondria promotes tau-mediated neurodegeneration and Alzheimer's disease-related tau phosphorylation via PAR-1** *PLoS Genet* **8**:e1002918 <https://doi.org/10.1371/journal.pgen.1002918> | Google Scholar
25. Lopez-Domenech G., Higgs N.F., Vaccaro V., Ros H., Arancibia-Carcamo I.L., MacAskill A.F., Kittler J.T (2016) **Loss of Dendritic Complexity Precedes Neurodegeneration in a Mouse Model with Disrupted Mitochondrial Distribution in Mature Dendrites** *Cell Rep* **17**:317–327 <https://doi.org/10.1016/j.celrep.2016.09.004> | Google Scholar
26. Guo X., Macleod G.T., Wellington A., Hu F., Panchumarthi S., Schoenfield M., Marin L., Charlton M.P., Atwood H.L., Zinsmaier K.E (2005) **The GTPase dMiro is required for axonal transport of mitochondria to Drosophila synapses** *Neuron* **47**:379–393 <https://doi.org/10.1016/j.neuron.2005.06.027> | Google Scholar
27. Glater E.E., Megeath L.J., Stowers R.S., Schwarz T.L (2006) **Axonal transport of mitochondria requires mltin to recruit kinesin heavy chain and is light chain independent** *J. Cell Biol* **173**:545–557 <https://doi.org/10.1083/jcb.200601067> | Google Scholar
28. Stowers R.S., Megeath L.J., Gorska-Andrzejak J., Meinertzhagen I.A., Schwarz T.L (2002) **Axonal transport of mitochondria to synapses depends on mltin, a novel Drosophila protein** *Neuron* **36**:1063–1077 [https://doi.org/10.1016/s0896-6273\(02\)01094-2](https://doi.org/10.1016/s0896-6273(02)01094-2) | Google Scholar
29. Iijima-Ando K., Hearn S.A., Shenton C., Gatt A., Zhao L., Iijima K (2009) **Mitochondrial mislocalization underlies Abeta42-induced neuronal dysfunction in a Drosophila model of Alzheimer's disease** *PLoS One* **4**:e8310 <https://doi.org/10.1371/journal.pone.0008310> | Google Scholar

30. Tonoki A., Kuranaga E., Tomioka T., Hamazaki J., Murata S., Tanaka K., Miura M (2009) **Genetic evidence linking age-dependent attenuation of the 26S proteasome with the aging process** *Mol. Cell. Biol* **29**:1095–1106 <https://doi.org/10.1128/MCB.01227-08> | Google Scholar
31. Klionsky D.J., Abdel-Aziz A.K., Abdelfatah S., Abdellatif M., Abdoli A., Abel S., Abeliovich H., Abildgaard M.H., Abudu Y.P., Acevedo-Arozena A., et al. (2021) **Guidelines for the use and interpretation of assays for monitoring autophagy (4th edition)(1)** *Autophagy* **17**:1–382 <https://doi.org/10.1080/15548627.2020.1797280> | Google Scholar
32. Bartlett B.J., Isakson P., Lewerenz J., Sanchez H., Kotzebue R.W., Cumming R.C., Harris G.L., Nezis I.P., Schubert D.R., Simonsen A., Finley K.D (2011) **p62, Ref(2)P and ubiquitinated proteins are conserved markers of neuronal aging, aggregate formation and progressive autophagic defects** *Autophagy* **7**:572–583 <https://doi.org/10.4161/auto.7.6.14943> | Google Scholar
33. Oka M., Suzuki E., Asada A., Saito T., Iijima K.M., Ando K. (2021) **Increasing neuronal glucose uptake attenuates brain aging and promotes life span under dietary restriction in *Drosophila*** *iScience* **24** <https://doi.org/10.1016/j.isci.2020.101979> | Google Scholar
34. Zia A., Pourbagher-Shahri A.M., Farkhondeh T., Samarghandian S (2021) **Molecular and cellular pathways contributing to brain aging** *Behav Brain Funct* **17** <https://doi.org/10.1186/s12993-021-00179-9> | Google Scholar
35. Haigis M.C., Yankner B.A (2010) **The aging stress response** *Mol. Cell* **40**:333–344 <https://doi.org/10.1016/j.molcel.2010.10.002> | Google Scholar
36. Wek R.C (2018) **Role of eIF2alpha Kinases in Translational Control and Adaptation to Cellular Stress** *Cold Spring Harb Perspect Biol* **10** <https://doi.org/10.1101/cshperspect.a032870> | Google Scholar
37. Currais A., Fischer W., Maher P., Schubert D (2017) **Intraneuronal protein aggregation as a trigger for inflammation and neurodegeneration in the aging brain** *FASEB J* **31**:5–10 <https://doi.org/10.1096/fj.201601184> | Google Scholar
38. Grimm A., Eckert A (2017) **Brain aging and neurodegeneration: from a mitochondrial point of view** *J. Neurochem* **143**:418–431 <https://doi.org/10.1111/jnc.14037> | Google Scholar
39. Bond S., Lopez-Lloreda C., Gannon P.J., Akay-Espinoza C., Jordan-Sciutto K.L (2020) **The Integrated Stress Response and Phosphorylated Eukaryotic Initiation Factor 2alpha in Neurodegeneration** *J. Neuropathol. Exp. Neurol* **79**:123–143 <https://doi.org/10.1093/jnen/nlz129> | Google Scholar
40. B'Chir W., Maurin A.C., Carraro V., Averous J., Jousse C., Muranishi Y., Parry L., Stepien G., Fafournoux P., Bruhat A (2013) **The eIF2alpha/ATF4 pathway is essential for stress-induced autophagy gene expression** *Nucleic Acids Res* **41**:7683–7699 <https://doi.org/10.1093/nar/gkt563> | Google Scholar
41. Salton G.D., Laurino C., Mega N.O., Delgado-Canedo A., Setterblad N., Carmagnat M., Xavier R.M., Cirne-Lima E., Lenz G., Henriques J.A.P., Laurino J.P (2017) **Deletion of eIF2beta lysine stretches creates a dominant negative that affects the translation and proliferation in human cell line: A tool for arresting the cell growth** *Cancer Biol Ther* **18**:560–570 <https://doi.org/10.1080/15384047.2017.1345383> | Google Scholar

42. Lee S., Nahm M., Lee M., Kwon M., Kim E., Zadeh A.D., Cao H., Kim H.J., Lee Z.H., Oh S.B., et al. (2007) **The F-actin-microtubule crosslinker Shot is a platform for Krasavietz-mediated translational regulation of midline axon repulsion** *Development* **134**:1767–1777 <https://doi.org/10.1242/dev.02842> | Google Scholar
43. Wu Z., Tantray I., Lim J., Chen S., Li Y., Davis Z., Sitron C., Dong J., Gispert S., Auburger G., et al. (2019) **MISTERMINATE Mechanistically Links Mitochondrial Dysfunction with Proteostasis Failure** *Mol. Cell* **75**:835–848 <https://doi.org/10.1016/j.molcel.2019.06.031> | Google Scholar
44. Sugiura A., Nagashima S., Tokuyama T., Amo T., Matsuki Y., Ishido S., Kudo Y., McBride H.M., Fukuda T., Matsushita N., et al. (2013) **MITOL regulates endoplasmic reticulum-mitochondria contacts via Mitofusin2** *Mol. Cell* **51**:20–34 <https://doi.org/10.1016/j.molcel.2013.04.023> | Google Scholar
45. Yonashiro R., Kimijima Y., Shimura T., Kawaguchi K., Fukuda T., Inatome R., Yanagi S (2012) **Mitochondrial ubiquitin ligase MITOL blocks S-nitrosylated MAP1B-light chain 1-mediated mitochondrial dysfunction and neuronal cell death** *Proc Natl Acad Sci U S A* **109**:2382–2387 <https://doi.org/10.1073/pnas.1114985109> | Google Scholar
46. Takeda K., Nagashima S., Shiiba I., Uda A., Tokuyama T., Ito N., Fukuda T., Matsushita N., Ishido S., Iwawaki T., et al. (2019) **MITOL prevents ER stress-induced apoptosis by IRE1alpha ubiquitylation at ER-mitochondria contact sites** *EMBO J* **38**:e100999 <https://doi.org/10.15252/embj.2018100999> | Google Scholar

Author information

Kanako Shinno[†]

Department of Biological Sciences, Graduate School of Science, Tokyo Metropolitan University, Tokyo, Japan
ORCID iD: [0009-0002-7114-7686](https://orcid.org/0009-0002-7114-7686)

[†]Department of Neuroscience and Pathobiology, Research Institute of Environmental Medicine, Nagoya University, Chikusa-Ku, Nagoya, Aichi, 464-8601, Japan

Yuri Miura

Research Team for Mechanism of Aging, Tokyo Metropolitan Institute for Geriatrics and Gerontology, Tokyo, Japan

Koichi M Iijima

Department of Alzheimer's Disease Research, National Center for Geriatrics and Gerontology, Ōbu, Japan, Department of Experimental Gerontology, Graduate School of Pharmaceutical Sciences, Nagoya City University, Nagoya, Japan

Emiko Suzuki

Department of Biological Sciences, Graduate School of Science, Tokyo Metropolitan University, Tokyo, Japan, Gene Network Laboratory, National Institute of Genetics and Department of Genetics, SOKENDAI, Mishima, Japan

Kanae Ando

Department of Biological Sciences, Graduate School of Science, Tokyo Metropolitan University, Tokyo, Japan, Department of Biological Sciences, School of Science, Tokyo

Metropolitan University, Tokyo, Japan
ORCID iD: [0000-0002-3956-276X](https://orcid.org/0000-0002-3956-276X)

For correspondence: k_ando@tmu.ac.jp

Editors

Reviewing Editor

Adam Frost

University of California, San Francisco (Adjunct), San Francisco, United States of America

Senior Editor

Pankaj Kapahi

Buck Institute for Research on Aging, Novato, United States of America

Reviewer #1 (Public review):

The study presents significant findings on the role of mitochondrial depletion in axons and its impact on neuronal proteostasis. It effectively demonstrates how the loss of axonal mitochondria and elevated levels of eIF2 β contribute to autophagy collapse and neuronal dysfunction. The use of *Drosophila* as a model organism and comprehensive proteome analysis adds robustness to the findings.

In this revision, the authors have responded thoughtfully to previous concerns. In particular, they have addressed the need for a quantitative analysis of age-dependent changes in eIF2 β and eIF2 α . By adding western blot data from multiple time points (7 to 63 days), they show that eIF2 β levels gradually increase until middle age, then decline. In *miton* knockdown flies, this pattern appears shifted, supporting the idea that mitochondrial defects may accelerate aging-related molecular changes. These additions clarify the temporal dynamics of eIF2 β and improve the overall interpretation.

Other updates include appropriate corrections to figures and quantification methods. The authors have also revised some of their earlier mechanistic claims, presenting a more cautious interpretation of their findings.

Overall, this work provides new insights into how mitochondrial transport defects may influence aging-related proteostasis through eIF2 β . The manuscript is now more convincing, and the revisions address the main points raised earlier. I find the updated version much improved.

<https://doi.org/10.7554/eLife.95576.4.sa2>

Reviewer #2 (Public review):

In the manuscript, the authors aimed to elucidate the molecular mechanism that explains neurodegeneration caused by the depletion of axonal mitochondria. In *Drosophila*, starting with siRNA depletion of *miton* and *Miro*, the authors attempted to demonstrate that the depletion of axonal mitochondria induces the defect in autophagy. From proteome analyses, the authors hypothesized that autophagy is impacted by the abundance of eIF2 β and the phosphorylation of eIF2 α . The authors followed up the proteome analyses by testing the effects of eIF2 β overexpression and depletion on autophagy. With the results from those experiments, the authors proposed a novel role of eIF2 β in proteostasis that underlies neurodegeneration derived from the depletion of axonal mitochondria, which they suggest accelerates age-dependent changes rather than increasing their magnitude.

Strong caution is necessary regarding the interpretation of translational regulation resulting from the milton KD. The effect of milton KD on translation appears subtle, if present at all, in the puromycin incorporation experiments in both the initial and revised versions. Additionally, the polysome profiling data in the revised manuscript lack the clear resolution for ribosomal subunits, monosomes, and polysomes that is typically expected in publications.

<https://doi.org/10.7554/eLife.95576.4.sa1>

Author response:

The following is the authors' response to the previous reviews

Reviewer #1 (Public review):

*The study presents significant findings on the role of mitochondrial depletion in axons and its impact on neuronal proteostasis. It effectively demonstrates how the loss of axonal mitochondria and elevated levels of eIF2 β contribute to autophagy collapse and neuronal dysfunction. The use of *Drosophila* as a model organism and comprehensive proteome analysis adds robustness to the findings.*

In this revision, the authors have responded thoughtfully to previous concerns. In particular, they have addressed the need for a quantitative analysis of age-dependent changes in eIF2 β and eIF2 α . By adding western blot data from multiple time points (7 to 63 days), they show that eIF2 β levels gradually increase until middle age, then decline. In milton knockdown flies, this pattern appears shifted, supporting the idea that mitochondrial defects may accelerate aging-related molecular changes. These additions clarify the temporal dynamics of eIF2 β and improve the overall interpretation.

Other updates include appropriate corrections to figures and quantification methods. The authors have also revised some of their earlier mechanistic claims, presenting a more cautious interpretation of their findings.

Overall, this work provides new insights into how mitochondrial transport defects may influence aging-related proteostasis through eIF2 β . The manuscript is now more convincing, and the revisions address the main points raised earlier. I find the updated version much improved.

Thank you so much for the review, insightful comments and encouragement. We appreciate it.

Reviewer #2 (Public review):

*In the manuscript, the authors aimed to elucidate the molecular mechanism that explains neurodegeneration caused by the depletion of axonal mitochondria. In *Drosophila*, starting with siRNA depletion of milton and Miro, the authors attempted to demonstrate that the depletion of axonal mitochondria induces the defect in autophagy. From proteome analyses, the authors hypothesized that autophagy is impacted by the abundance of eIF2 β and the phosphorylation of eIF2 α . The authors followed up the proteome analyses by testing the effects of eIF2 β overexpression and depletion on autophagy. With the results from those experiments, the authors proposed a novel role of eIF2 β in proteostasis that underlies neurodegeneration derived from the depletion of axonal mitochondria, which they suggest accelerates age-dependent changes rather than increasing their magnitude.*

Strong caution is necessary regarding the interpretation of translational regulation resulting from the milton KD. The effect of milton KD on translation appears subtle, if present at all, in the puromycin incorporation experiments in both the initial and revised versions. Additionally, the polysome profiling data in the revised manuscript lack the clear resolution for ribosomal subunits, monosomes, and polysomes that is typically expected in publications.

Thank you so much for the review and insightful comments. We appreciate it.

Reviewer #2 (Recommendations for the authors):

The revised manuscript demonstrates many improvements. The authors have provided a more comprehensive data set and a more detailed description of their results. Furthermore, their explanation of the Integrated Stress Response (ISR) has been corrected, and this correction is reflected in the data interpretation.

As in the public review, I maintained my emphasis on the weakness of the claim on suppressed global translation, since the data are the same in the initial and the revised versions.

Thank you for your review. We understand that further studies will be needed to elucidate the roles on mitochondrial distribution in global translation profile. We will keep working on it.

A few suggestions for minor corrections.

(1) The order of figures in the revised version is disorganized.

Thank you for pointing it out. We corrected the order.

(2) In Figure 1A, mitochondria is bound by milton, and kinesin is bound by Miro. Their roles should be opposite.

Thank you for pointing it out, and we are sorry for the oversight. We corrected it.

<https://doi.org/10.7554/eLife.95576.4.sa0>